

Project Data Page:



Table of Contents

Cover Page	1
Title Page	2
Vessel Design Summary:	3
Nozzle Summary:	6
Nozzle Schedule:	7
Bill of Material:	9
MDMT Summary:	10
Warnings and Errors:	12
Input Echo:	13
XY Coordinate Calculations:	22
Internal Pressure Calculations:	23
Element: EH-1	24
Element: Shell-3	26
Element: SHELL-2	28
Element: SHELL-3	30
Element: EH-2	32
Shop Pressure Test Results:	34
Field Pressure Test Results:	36
External Pressure Calculations:	38
Element and Detail Weights:	39
Nozzle Flange MAWP:	41
Wind Load Calculation:	43
Earthquake Load Calculation:	45
Center of Gravity Calculation:	47
Saddle Calcs: Operating Case:	48
Saddle Calcs: Test Case:	62
Nozzle Calcs.: M1-24''	76
Nozzle Calcs.: K2A-2''	82
Nozzle Calcs.: K2B-2''	85
Nozzle Calcs.: N3-4''	88
Nozzle Calcs.: N4-4''	102
Nozzle Calcs.: K3-2''	116
Nozzle Calcs.: K4-2''	119
Nozzle Calcs.: N5-4''	122
Nozzle Calcs.: N6-4''	136
Nozzle Calcs.: K1A-2''	150
Nozzle Calcs.: K1B-2''	153
Nozzle Calcs.: N1-20''	156
Nozzle Calcs.: N2-4''	170
Nozzle Calcs.: M2-24''	176



Expansion vessel (AMONIA PRODUCTION)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A162-V-XX02-Expansion Vessel (Ammonia production & Other Area)-7%.pvdb

Date of Analysis : Sep 17, 2026 2:01pm

PV Elite 28, January 2026



Customer : by:
item : date:
Item No : S/O:

Design Internal Pressure : Temp. :
Design External Pressure : Temp. :

Head Matl.:	Corr. All.:	JE :
Shell Matl.:	Corr. All.:	JE :
Flange Matl.:	Corr. All.:	JE :
Pipe Matl.:	Corr. All.:	JE :
Cone Matl.:	Corr. All.:	JE :

Radiographic Requirements :

Post Weld Heat Treat:

Wind Specification : V : Exp.: Imp :

Seismic : Zone : Cat.:

Results ...

Basic Flange Class: Rating : Ambient:

MAWP (Corroded) : limited by :
MAP (New & Cold): limited by :
Min. Design Metal Tmp: without Impacts
Charpy Impact Reqmnts.:

Stiffening Rings Required :

hydrotest Requirement :

Notes ...

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 3290.000 mm ID	
Vessel Design Length, Tangent to Tangent	9860.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	1.000 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	1 MPa
Shop Test Pressure	1.347 MPa
Field Test Pressure	1.347 MPa
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
EH-1	1.026	0.00	1.000	3.0000	No	No
Shell-3	1.026	0.00	1.000	3.0000	N/A	No
SHELL-2	1.026	0.00	1.000	3.0000	N/A	No
SHELL-3	1.026	0.00	1.000	3.0000	N/A	No
EH-2	1.026	0.00	1.000	3.0000	No	No

Liquid Level: 2630.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000**Element Types and Properties:**

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	20.0	18.0	17.4	...	0.85	0.85
Cylinder	3050.0	3000.0	18.0	18.0	17.5	...	0.85	0.85
Cylinder	6050.0	3000.0	18.0	18.0	17.5	...	0.85	0.85
Cylinder	9810.0	3760.0	18.0	18.0	17.5	...	0.85	0.85
Ellipse	9860.0	50.0	20.0	18.0	17.4	...	0.85	0.85



Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-4''	WRC-297	0.826	Shell	Passed
N4-4''	WRC-297	0.826	Shell	Passed
N5-4''	WRC-297	0.826	Shell	Passed
N6-4''	WRC-297	0.826	Shell	Passed
N1-20''	WRC-297	0.993	Shell	Passed

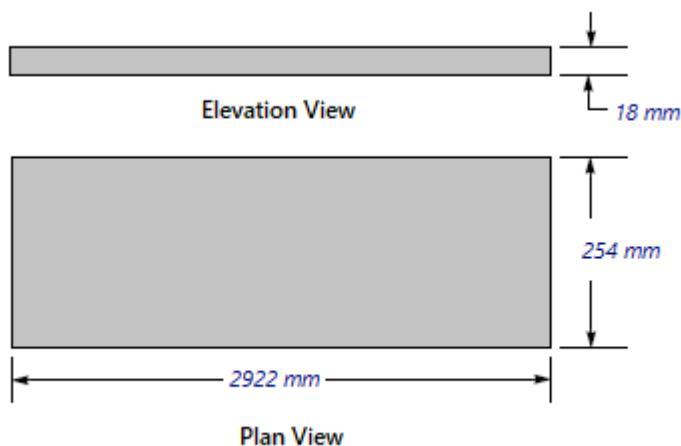
Foundation/Support Summary

Saddle Parameters:

Saddle Width	228.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	2300.000 mm
Wear Pad Width	280.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1000.000 mm

Baseplate Length	2922.000 mm
Baseplate Thickness	18.000 mm
Baseplate Width	254.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	13.000 mm
Web Thickness	16.000 mm
Height of Center Web	619.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	665187.88 N
Maximum Transverse Saddle Shear Load	50326.26 N
Maximum Longitudinal Saddle Shear Load	100652.52 N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	631050.06 N
Maximum Transverse Saddle Shear Load	12204.29 N
Maximum Longitudinal Saddle Shear Load	9698.89 N

Weights:

Fabricated - Bare w/o removable internals	23365.7 kgm
Shop Test - Fabricated + water (full)	116470.7 kgm

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User



FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 5 of 181

Vessel Design Summary: Step: 37 2:01pm Sep 17,2026



HEXAGON

Shipping	- Fab. + rem. intls.+ shipping app.	23365.7	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	29548.0	kgm
Empty	- Fab. + intls. + details + wghts.	29548.0	kgm
Operating	- Empty + operating liquid (no ca)	109729.7	kgm
Field Test	- Empty weight + test medium (full)	122652.9	kgm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Nozzle Calculation Summary (MPa & mm):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1-24''	1	...	...	...		OK	Passed
K2A-2''	1	...	...	OK	6.42	OK	No Calc[*]
K2B-2''	1	...	...	OK	6.42	OK	No Calc[*]
N3-4''	1	...	...	OK	8.26	OK	Passed
N4-4''	1	...	...	OK	8.26	OK	Passed
K3-2''	1	...	...	OK	6.42	OK	No Calc[*]
K4-2''	1	...	...	OK	6.42	OK	No Calc[*]
N5-4''	1	...	...	OK	8.26	OK	Passed
N6-4''	1	...	...	OK	8.26	OK	Passed
K1A-2''	1	...	...	OK	6.42	OK	No Calc[*]
K1B-2''	1	...	...	OK	6.42	OK	No Calc[*]
N1-20''	1	...	...	OK	11.33	OK	Passed
N2-4''	7	...	...	OK	5.16	OK	Passed
M2-24''	1	...	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : N1-20''
 Minimum MAWP Shells/Flanges : 1 Element : SHELL-3
 Minimum MAPnc Shells/Flanges : 1 Element : SHELL-3

Computed Vessel M.A.W.P. : 1 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1-24''	0.000	0.000	1183.200
20	K2A-2''	550.000	90.000	80.800
20	K2B-2''	550.000	270.000	80.800
20	N3-4''	1550.000	90.000	201.662
20	N4-4''	2950.000	90.000	201.662
30	K3-2''	4250.000	90.000	80.800
30	K4-2''	5450.000	90.000	80.800
40	N5-4''	6550.000	90.000	201.662
40	N6-4''	7250.001	90.000	201.662
40	K1A-2''	9250.000	90.000	80.800
40	K1B-2''	9250.000	270.000	80.800
40	N1-20''	6850.000	270.000	956.000
40	N2-4''	0.000	180.000	201.662
50	M2-24''	0.000	0.000	1183.200

The nozzle spacing is computed by the following:

$$= \sqrt{(l_l^2 + l_c^2)} \text{ where}$$

l_l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2A-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	318.20	150
K2B-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	318.20	150
K3-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	318.20	150
K4-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	318.20	150
K1A-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	318.20	150
K1B-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	218.20	150
N3-4''	4.000 in	120	WNF	4.500	11.125	200.00	18.00	318.99	300
N4-4''	4.000 in	120	WNF	4.500	11.125	200.00	18.00	318.99	300
N5-4''	4.000 in	120	WNF	4.500	11.125	200.00	18.00	318.99	300
N6-4''	4.000 in	120	WNF	4.500	11.125	200.00	18.00	318.99	300
N2-4''	4.000 in	120	WNF	4.500	11.125	200.00	18.00	425.02	150
N1-20''	20.000 in	Actual	WNF	20.000	18.000	1000.00	18.00	437.73	150
M1-24''	24.000 in	Actual	WNF	24.000	12.000	1000.00	18.00	333.65	150
M2-24''	24.000 in	Actual	WNF	24.000	12.000	1000.00	18.00	333.65	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2A-2''	SA-106 B	18.000	10.000	...	...	...
K2B-2''	SA-106 B	18.000	10.000	...	...	...
K3-2''	SA-106 B	18.000	10.000	...	...	...
K4-2''	SA-106 B	18.000	10.000	...	...	...
K1A-2''	SA-106 B	18.000	10.000	...	...	...
K1B-2''	SA-106 B	18.000	10.000	...	...	...
N3-4''	SA-106 B	18.000	10.000	14.000	18.000	...
N4-4''	SA-106 B	18.000	10.000	14.000	18.000	...
N5-4''	SA-106 B	18.000	10.000	14.000	18.000	...
N6-4''	SA-106 B	18.000	10.000	14.000	18.000	...
N2-4''	SA-106 B	18.000	10.000	14.000	18.000	...
N1-20''	SA-106 B	18.000	10.000	14.000	18.000	...
M1-24''	SA-516 70	18.000	10.000	18.000	18.000	...
M2-24''	SA-516 70	18.000	10.000	14.000	18.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2A-2''	550.000	90.0	300.00	0.00	Shell-3
K2B-2''	550.000	270.0	300.00	0.00	Shell-3
K3-2''	4250.000	90.0	300.00	0.00	SHELL-2
K4-2''	5450.000	90.0	300.00	0.00	SHELL-2
K1A-2''	9250.000	90.0	300.00	0.00	SHELL-3
K1B-2''	9250.000	270.0	200.00	0.00	SHELL-3
N3-4''	1550.000	90.0	300.00	0.00	Shell-3
N4-4''	2950.000	90.0	300.00	0.00	Shell-3

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User

FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 8 of 181

Nozzle Schedule: Step: 34 2:01pm Sep 17,2026



HEXAGON

N5-4''	6550.000	90.0	300.00	0.00	SHELL-3
N6-4''	7250.000	90.0	300.00	0.00	SHELL-3
N2-4''	6250.000	180.0	400.00	0.00	N1-20''
N1-20''	6850.000	270.0	400.00	0.00	SHELL-3
M1-24''	...	0.0	300.00	0.00	EH-1
M2-24''	...	0.0	300.00	0.00	EH-2

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User

FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 9 of 181

Bill of Material: Step: 33 2:01pm Sep 17,2026



HEXAGON

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 3290.0mm ID X 50.0mm	SA-516 70
2	CYLINDER: 18.0mm THK X 3290.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 18.0mm THK X 3290.0mm ID X 3760.0mm	SA-516 70
2	SADDLE: 228.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1000mm X 3326mm WIDE	
1	LADDER: 4000.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
EH-1	[10]	B	-11	-19	-29	0.844	18.000	0.85	No
EH-1	[7]	B	-7	-21	-29	0.750	20.000	0.85	No
Shell-3	[8]	B	-11	-19	-29	0.850	18.000	0.85	No
SHELL-2	[8]	B	-11	-19	-29	0.850	18.000	0.85	No
SHELL-3	[8]	B	-11	-19	-29	0.850	18.000	0.85	No
EH-2	[10]	B	-11	-19	-29	0.844	18.000	0.85	No
EH-2	[7]	B	-7	-21	-29	0.750	20.000	0.85	No
Left Saddle	[24]	B	-29	-37	-29	0.850	10.000	0.85	No
Left Saddle	[24]	B	-29	-37	-29	0.850	10.000	0.85	No
M1-24''	[1]	B	-11	-29	-29	0.749	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-46					
K2A-2''	[1]	B	-29	-104	-29	0.032	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-47					
K2B-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
N3-4''	[1]	B	-11	-29	-29	0.829	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-4''	[1]	B	-11	-29	-29	0.829	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2''	[1]	B	-29	-104	-29	0.032	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-47					
K4-2''	[1]	B	-29	-104	-29	0.032	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-47					
N5-4''	[1]	B	-11	-29	-29	0.829	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-4''	[1]	B	-11	-29	-29	0.829	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2''	[1]	B	-29	-104	-29	0.032	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-47					
K1B-2''	[1]	B	-29	-104	-29	0.033	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
N1-20''	[1]	B	-11	-29	-29	0.849	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-45					
N2-4''	[1]	B	-8	-104	-29	0.076	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-45					
M2-24''	[1]	B	-11	-29	-29	0.749	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-46					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -7 -19

Required Minimum Design Metal Temperature -10.0 °C
Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)



[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations

[21] - Body Flange Bolting Material

[22] - Nozzle Flange Bolting Material

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Units used in this Analysis (SIASME):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Expansion vessel (AMONIA PRODUCTION)

Design Internal Pressure (for Hydrotest)	1	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-4	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13		IP+WE+EW	
Load Case 14		IP+WF+CW	
Load Case 15		IP+VO+OW	
Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-22	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	180	km/hr
Surface Roughness Category	D: Flat, unobstructed		
Type of Surface		0	
Base Elevation		0	mm
Percent Wind for Hydrotest		33.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm
Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg
M.A.W.P + Static Head to Nozzle		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes



Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	EH-1	
Distance "FROM" to "TO"	50	mm
Inside Diameter	3290	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2630	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	10	
Detail Type	Nozzle	
Detail ID	M1-24''	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4254.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3290	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	950	mm
Width of Saddle	228	mm
Height of Saddle at Bottom	2300	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	280	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	20	
Detail Type	Platform	
Detail ID	PF	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	14931300	mm ²
Platform Weight	56657	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	3326	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.3	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	9860	mm

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2630	mm
Liquid Density	0.0009984	kg/cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2A-2''	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	60.842	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
-------------------	----	--



Detail Type	Nozzle
Detail ID	K2B-2''
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	60.842 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	20
Detail Type	Nozzle
Detail ID	N3-4''
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	275.67 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	20
Detail Type	Nozzle
Detail ID	N4-4''
Dist. from "FROM" Node / Offset dist	2900 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	275.67 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	20
Detail Type	Weight
Detail ID	Cleat weight
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	0 mm

Element From Node	20
Detail Type	Weight
Detail ID	LL-1
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	3039.9 N
Offset from Element Centerline	1700 mm

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	SHELL-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3290 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2630	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-2''	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	60.842	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K4-2''	
Dist. from "FROM" Node / Offset dist	2400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	60.842	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/5	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	SHELL-3	
Distance "FROM" to "TO"	3760	mm
Inside Diameter	3290	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2810	mm
Width of Saddle	228	mm
Height of Saddle at Bottom	2300	mm
Saddle Contact Angle	120.0	



Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	280	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2630	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N5-4''	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	275.67	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N6-4''	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	275.67	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1A-2''	
Dist. from "FROM" Node / Offset dist	3200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	60.842	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1B-2''	
Dist. from "FROM" Node / Offset dist	3200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	51.101	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	40
Detail Type	Nozzle
Detail ID	N1-20''
Dist. from "FROM" Node / Offset dist	800 mm
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	2595.1 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	N2-4''
Dist. from "FROM" Node / Offset dist	200 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	230.41 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Weight
Detail ID	LL-2
Dist. from "FROM" Node / Offset dist	2760 mm
Miscellaneous Weight	3039.9 N
Offset from Element Centerline	1700 mm

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	EH-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	3290 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2630 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	M2-24''
Dist. from "FROM" Node / Offset dist	0 mm

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User

FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 21 of 181

Input Echo: Step: 1 2:01pm Sep 17,2026



HEXAGON

Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	4254.9 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	EH-1	50	...	50	...
	Shell-3	3050	...	3000	...
	SHELL-2	6050	...	3000	...
	SHELL-3	9810	...	3760	...
	EH-2	9860	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	EH-1	1.0258	20	3	3290	117.3	1
	Shell-3	1.0258	18	3	3290	117.3	2
	SHELL-2	1.0258	18	3	3290	117.3	3
	SHELL-3	1.0258	18	3	3290	117.3	4
	EH-2	1.0258	20	3	3290	117.3	5

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	EH-1	1	1.04344	1.2821	18	17.39	1
	Shell-3	1	1.03603	1.27513	18	17.4883	2
	SHELL-2	1	1.03603	1.27513	18	17.4883	3
	SHELL-3	1	1.03603	1.27513	18	17.4883	4
	EH-2	1	1.04344	1.2821	18	17.39	5

Minimum

1

1

MAWP: 1 MPa, limited by: SHELL-3.

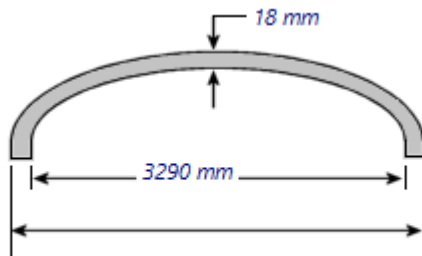
Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-1

Elliptical Head Design Information:					
Design Pressure	P	1.026 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3290.000 mm	Actual Stress		132.4 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		12543278.0 mm²	SG of Contents		1.000
Element Volume		5087.5 ltr	Weight of Contents		44643.1 N
Empty Weight		20722.2 N	Operating Weight		65365.3 N
			Comp. All. Ope. Stress		106.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 41.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.562 mm

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P)$ Appendix 1-4(c)
 $= (1.026 \cdot 3296.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 1.026)$
 $= 14.3900 + 3.0000 = 17.3900 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 MPa
 $= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 138.0 \cdot 0.85 \cdot 15.0) / (0.998 \cdot 3296.0 + 0.2 \cdot 15.0)$
 $= 1 - 0 = 1 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 138.0 \cdot 0.85 \cdot 18.0) / (1.0 \cdot 3290.0 + 0.2 \cdot 18.0)$
 $= 1 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
 $= (1.026 \cdot (0.998 \cdot 3296.0 + 0.2 \cdot 15.0)) / (2 \cdot 0.85 \cdot 15.0)$
 $= 132.391 \text{ MPa}$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (1.026 \cdot 1648.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1.026) + 3.0$$

$$= 17.488 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (1648.0 + 0.6 \cdot 17.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3296.0 / (2 \cdot 825.5)))^2 / 6$$

$$= 0.997579$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.635 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18.0, tr = 14.896, c = 3.0 mm, E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.844$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20.0, tr = 15.0, c = 3.0 mm, E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.062 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.750	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-21 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 41.125 mm

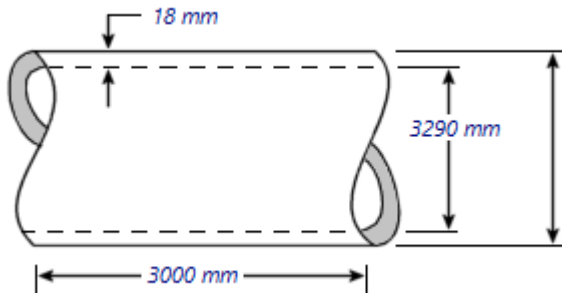
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.562 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.026 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3290.000 mm	Actual Stress	133.3 MPa	
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		31346812.0 mm²	SG of Contents	1.000	
Element Volume		25508.3 ltr	Weight of Contents	214234.4 N	
Empty Weight		42651.2 N	Operating Weight	256885.6 N	
			Comp. All. Ope. Stress	106.7 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.026 \cdot 1648.0) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 1.026) \\
 &= 14.4883 + 3.0000 = 17.4883 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (1648.0 + 0.6 \cdot 15.0) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 18.0) / (1645.0 + 0.6 \cdot 18.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.026 \cdot (1648.0 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 133.315 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.544 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 18.0, tr = 15.0, c = 3.0 mm, E* = 0.85
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

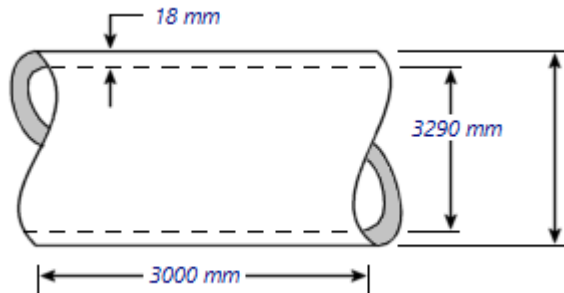
Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -19 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.062 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-19 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.026 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3290.000 mm	Actual Stress	133.3 MPa	
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		31346812.0 mm²	SG of Contents	1.000	
Element Volume		25508.3 ltr	Weight of Contents	214234.4 N	
Empty Weight		42651.2 N	Operating Weight	256885.6 N	
			Comp. All. Ope. Stress	106.7 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.026 \cdot 1648.0) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 1.026) \\
 &= 14.4883 + 3.0000 = 17.4883 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (1648.0 + 0.6 \cdot 15.0) \\
 &= 1 - 0 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 18.0) / (1645.0 + 0.6 \cdot 18.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.026 \cdot (1648.0 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 133.315 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.544 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 18.0, tr = 15.0, c = 3.0 mm, E* = 0.85
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

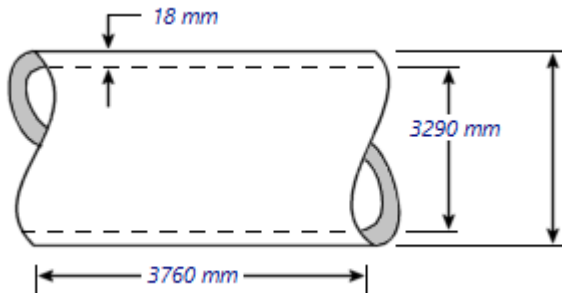
Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -19 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.062 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-19 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.026 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3760.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3290.000 mm	Actual Stress		133.3 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		39288000.0 mm²	SG of Contents		1.000
Element Volume		31970.3 ltr	Weight of Contents		268507.2 N
Empty Weight		53456.2 N	Operating Weight		321963.3 N
			Comp. All. Ope. Stress		106.7 MPa
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.026 \cdot 1648.0) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 1.026)$$

$$= 14.4883 + 3.0000 = 17.4883 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 15.0) / (1648.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 18.0) / (1645.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.026 \cdot (1648.0 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0)$$

$$= 133.315 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.544 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 18.0, tr = 15.0, c = 3.0 mm, E* = 0.85
Thickness Ratio = $tr * E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

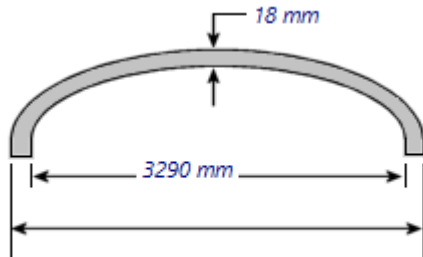
Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -19 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.062 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-19 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-2

Elliptical Head Design Information:					
Design Pressure	P	1.026 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3290.000 mm	Actual Stress		132.4 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		12543278.0 mm²	SG of Contents		1.000
Element Volume		5087.5 ltr	Weight of Contents		44643.1 N
Empty Weight		20722.2 N	Operating Weight		65365.3 N
			Comp. All. Ope. Stress		106.7 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 41.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.562 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.026 \cdot 3296.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 1.026)$$

$$= 14.3900 + 3.0000 = 17.3900 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 15.0) / (0.998 \cdot 3296.0 + 0.2 \cdot 15.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 18.0) / (1.0 \cdot 3290.0 + 0.2 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.026 \cdot (0.998 \cdot 3296.0 + 0.2 \cdot 15.0)) / (2 \cdot 0.85 \cdot 15.0)$$

$$= 132.391 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (1.026 \cdot 1648.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1.026) + 3.0$$

$$= 17.488 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.026 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (1648.0 + 0.6 \cdot 17.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3296.0 / (2 \cdot 825.5)))^2 / 6$$

$$= 0.997579$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.635 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18.0, tr = 14.896, c = 3.0 mm, E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.844$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20.0, tr = 15.0, c = 3.0 mm, E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.062 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.750	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-21 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 41.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 20.562 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.347 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.625 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.140 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.036 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 1.036 • 1.0

= 1.347 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	148.4	234.0	0.634	1.38
Shell-3	149.2	234.0	0.638	1.38
SHELL-2	149.2	234.0	0.638	1.38
SHELL-3	149.2	234.0	0.638	1.38
EH-2	148.4	234.0	0.634	1.38

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
K2A-2''	Nozzle	118.00	118.00	1.000
K2B-2''	Nozzle	118.00	118.00	1.000
N3-4''	Nozzle	118.00	118.00	1.000
N3-4''	Pad	138.00	138.00	1.000
N4-4''	Nozzle	118.00	118.00	1.000
N4-4''	Pad	138.00	138.00	1.000
K3-2''	Nozzle	118.00	118.00	1.000
K4-2''	Nozzle	118.00	118.00	1.000
N5-4''	Nozzle	118.00	118.00	1.000
N5-4''	Pad	138.00	138.00	1.000
N6-4''	Nozzle	118.00	118.00	1.000
N6-4''	Pad	138.00	138.00	1.000
K1A-2''	Nozzle	118.00	118.00	1.000
K1B-2''	Nozzle	118.00	118.00	1.000
N1-20''	Nozzle	118.00	118.00	1.000
N1-20''	Pad	138.00	138.00	1.000
N2-4''	Nozzle	118.00	118.00	1.000
N2-4''	Pad	138.00	138.00	1.000
M2-24''	Nozzle	138.00	138.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000



SHELL-3	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24''	34.48	234.00	0.147
K2A-2''	2.95	216.00	0.014
K2B-2''	2.95	216.00	0.014
N3-4''	7.54	216.00	0.035
N4-4''	7.54	216.00	0.035
K3-2''	2.95	216.00	0.014
K4-2''	2.95	216.00	0.014
N5-4''	7.54	216.00	0.035
N6-4''	7.54	216.00	0.035
K1A-2''	2.95	216.00	0.014
K1B-2''	2.95	216.00	0.014
N1-20''	18.91	216.00	0.088
N2-4''	7.54	216.00	0.035
M2-24''	34.48	234.00	0.147

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.347 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.625 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.140 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 MPa
App 27-4	= M.A.W.P.	1.036 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 1.036 • 1.0

= 1.347 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	178.0	234.0	0.761	1.38
Shell-3	179.2	234.0	0.766	1.38
SHELL-2	179.2	234.0	0.766	1.38
SHELL-3	179.2	234.0	0.766	1.38
EH-2	178.0	234.0	0.761	1.38

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24''	Nozzle	138.00	138.00	1.000
M1-24''	Pad	138.00	138.00	1.000
K2A-2''	Nozzle	118.00	118.00	1.000
K2B-2''	Nozzle	118.00	118.00	1.000
N3-4''	Nozzle	118.00	118.00	1.000
N3-4''	Pad	138.00	138.00	1.000
N4-4''	Nozzle	118.00	118.00	1.000
N4-4''	Pad	138.00	138.00	1.000
K3-2''	Nozzle	118.00	118.00	1.000
K4-2''	Nozzle	118.00	118.00	1.000
N5-4''	Nozzle	118.00	118.00	1.000
N5-4''	Pad	138.00	138.00	1.000
N6-4''	Nozzle	118.00	118.00	1.000
N6-4''	Pad	138.00	138.00	1.000
K1A-2''	Nozzle	118.00	118.00	1.000
K1B-2''	Nozzle	118.00	118.00	1.000
N1-20''	Nozzle	118.00	118.00	1.000
N1-20''	Pad	138.00	138.00	1.000
N2-4''	Nozzle	118.00	118.00	1.000
N2-4''	Pad	138.00	138.00	1.000
M2-24''	Nozzle	138.00	138.00	1.000
M2-24''	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000



SHELL-3	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1-24''	34.48	234.00	0.147
K2A-2''	2.95	216.00	0.014
K2B-2''	2.95	216.00	0.014
N3-4''	7.54	216.00	0.035
N4-4''	7.54	216.00	0.035
K3-2''	2.95	216.00	0.014
K4-2''	2.95	216.00	0.014
N5-4''	7.54	216.00	0.035
N6-4''	7.54	216.00	0.035
K1A-2''	2.95	216.00	0.014
K1B-2''	2.95	216.00	0.014
N1-20''	18.91	216.00	0.088
N2-4''	7.54	216.00	0.035
M2-24''	34.48	234.00	0.147

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	3326	15	0.00062638	62.6237
20	30	10408.3	3326	15	0.00012333	12.3304
30	40	10408.3	3326	15	0.00012333	12.3304
40	50	10408.3	3326	15	0.00012333	12.3304
50	60	No Calc	3326	15	0.00062638	62.6237

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	18	No Calc	...	0.31381
20	30	18	No Calc	...	0.074146
30	40	18	No Calc	...	0.074146
40	50	18	No Calc	...	0.074146
50	60	18	No Calc	...	0.31381

Minimum

0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	10408.3	No Calc	No Calc	No Calc
30	40	10408.3	No Calc	No Calc	No Calc
40	50	10408.3	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Element and Detail Weights:**

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	2113.22	5087.48	1796.23	5114.59	147.925
20	30	4349.5	25508.3	3627.87	25601.4	304.465
30	40	4349.5	25508.3	3627.87	25601.4	304.465
40	50	5451.37	31970.3	4546.93	32087.1	381.596
50	60	2113.22	5087.48	1796.23	5114.59	147.925
Total		18376	93161.81	15395	93518.99	1286

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	4552.63	-274.167	330	...	Liquid 10
10	Nozl	464.284	-822.5	...	...	M1-24''
20	Sadl	793.294	950	1958.5	...	Left Saddle
20	Plat	6182.24	4930	1739.2	...	PF
20	Liqd	21847.3	1500	330	...	Liquid 20
20	Nozl	6.63894	500	1670.4	...	K2A-2''
20	Nozl	6.63894	500	-1670.4	...	K2B-2''
20	Nozl	30.0803	1500	1702.15	...	N3-4''
20	Nozl	30.0803	2900	1702.15	...	N4-4''
20	Wght	100	1500	...	...	Cleat weight
20	Wght	310	1000	1472.24	850	LL-1
30	Liqd	21847.3	1500	330	...	Liquid 30
30	Nozl	6.63894	1200	1670.4	...	K3-2''
30	Nozl	6.63894	2400	1670.4	...	K4-2''
40	Sadl	793.294	2810	1958.5	...	Left Saddle
40	Liqd	27381.9	1880	330	...	Liquid 40
40	Nozl	30.0803	500	1702.15	...	N5-4''
40	Nozl	30.0803	1200	1702.15	...	N6-4''
40	Nozl	6.63894	3200	1670.4	...	K1A-2''
40	Nozl	5.57596	3200	-1670.4	...	K1B-2''
40	Nozl	283.167	800	-1899	...	N1-20''
40	Nozl	25.1414	200	...	-1702.15	N2-4''
40	Wght	310	2760	1472.24	850	LL-2
50	Liqd	4552.63	324.167	330	...	Liquid 50
50	Nozl	464.284	872.5	...	...	M2-24''

Total Weight of Each Detail Type:

Saddles	1586.6
Platforms	6182.2
Liquid	80181.8
Nozzles	1396.0
Weights	720.0

Sum of the Detail Weights 90066.6 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	19663.2	19663.2	19663.2	19663.2	19663.2	19663.2
Saddles	1586.6	1586.6	1586.6	1586.6	1586.6	1586.6
Nozzles	1396.0	1396.0	1396.0	1396.0	1396.0	1396.0
Wld Weights	720.0	720.0	720.0	720.0	720.0	720.0
Platforms	...	...	...	6182.2	6182.2	6182.2
Ope. Liquid	...	...	...	...	...	80181.8
Test Liquid	...	93104.9	...	...	...	...



Totals	23365.7	116470.7	23365.7	29548.0	29548.0	109729.7
--------	---------	----------	---------	---------	---------	----------

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	23365.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	116470.7 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	23365.7 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	29548.0 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	29548.0 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	109729.7 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	106539.4 kgm
Field Test Wt.	- Empty weight + test medium (full)	122652.9 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	12543278	12249957
20	30	31346812	31007518
30	40	31346812	31007518
40	50	39288000	38862756
50	60	12543278	12249957

Total	127068192.0	125377696.0 mm ²
Total	127.1	125.4 m ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Nozzle Flange MAWP Results: (MPa & °C)**

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M1-24''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K2A-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K2B-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N3-4''	4.76	5.10	85	300	GR 1.1	3.646	3.499	4.763	3.499
N4-4''	4.76	5.10	85	300	GR 1.1	3.646	3.499	4.763	3.499
K3-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K4-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N5-4''	4.76	5.10	85	300	GR 1.1	3.646	3.499	4.763	3.499
N6-4''	4.76	5.10	85	300	GR 1.1	3.646	3.499	4.763	3.499
K1A-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
K1B-2''	1.81	1.96	85	150	GR 1.1	...	...	...	...
N1-20''	1.81	1.96	85	150	GR 1.1	1.726	1.812	1.812	0.992
N2-4''	1.81	1.96	85	150	GR 1.1	...	...	...	...
M2-24''	1.81	1.96	85	150	GR 1.1	...	...	...	...
Min Rating	1.812	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3-4''	138.176	0.50	1764938	3600	1.036	3.499	Pass
N4-4''	138.176	0.50	1764938	3600	1.036	3.499	Pass
N5-4''	138.176	0.50	1764938	3600	1.036	3.499	Pass
N6-4''	138.176	0.50	1764938	3600	1.036	3.499	Pass
N1-20''	551.688	1.20	52502400	31999	1.036	1.812	Pass

Evaluating Flange on Nozzle [N3-4'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1764938) + 4 \cdot \text{abs}(3600) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot ((4.763 - 1.036) + .500 \cdot 4.763)$$

$$= 0.30230E+08 \leq 0.50647E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-4'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 30229554.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 3.499)$$

$$= 3.499 \text{ MPa}$$

Evaluating Flange on Nozzle [N4-4'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1764938) + 4 \cdot \text{abs}(3600) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot ((4.763 - 1.036) + .500 \cdot 4.763)$$

$$= 0.30230E+08 \leq 0.50647E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-4'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 30229554.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 3.499)$$

$$= 3.499 \text{ MPa}$$



Evaluating Flange on Nozzle [N5-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1764938) + 4 \cdot \text{abs}(3600) 138.176 \leq \pi \cdot 138.176^3 [(4.763 - 1.036) + .500 \cdot 4.763]$$

$$= 0.30230E+08 \leq 0.50647E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N5-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 30229554.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 3.499)$$

$$= 3.499 \text{ MPa}$$

Evaluating Flange on Nozzle [N6-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1764938) + 4 \cdot \text{abs}(3600) 138.176 \leq \pi \cdot 138.176^3 [(4.763 - 1.036) + .500 \cdot 4.763]$$

$$= 0.30230E+08 \leq 0.50647E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 30229554.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.763, 3.499)$$

$$= 3.499 \text{ MPa}$$

Evaluating Flange on Nozzle [N1-20"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(52502400) + 4 \cdot \text{abs}(31999) 551.688 \leq \pi \cdot 551.688^3 [(1.812 - 1.036) + 1.200 \cdot 1.812]$$

$$= 0.91068E+09 \leq 0.15570E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-20" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.812, 1.200 \cdot 1.812 + 1.812 - 910679936.0 / (\pi \cdot 551.688^3))$$

$$= \min(1.812, 2.261)$$

$$= 1.812 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.974) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.912)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.738
Structure Height to Diameter ratio	3.272

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 2300.0 + 0.0 = 2300.0$ mm
 $= 7.546$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
Because z (7.546 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf} [1.587] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.738 \cdot 25.781$$

$$= 536.0 \text{ lbs.} [2384.2] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
EH-1	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-3	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-2	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-3	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
EH-2	2300.0	0.000	0.000	0.000	1.035	1.000	1.587

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PF	14931300.00	2300.00	1.59	23697.44	1.30

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Cleat weight	0.00	2300.00	1.59	0.00
LL-1	0.00	3772.24	1.63	0.00
LL-2	0.00	3772.24	1.63	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	2300	3991.2	2395142	1.58714	2622.58
20	30	2300	3991.2	11973601	1.58714	39177.8
30	40	2300	3991.2	11973601	1.58714	13110.6
40	50	2300	3991.2	15006912	1.58714	16431.9
50	60	2300	3991.2	2395142	1.58714	2622.58

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(12.94290.75)$
 = 0.136 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.136$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 1029167 \cdot 1.5$
 = 100652.508 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 1029167$
 = 44734.4 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	1645	147024	14378.9
20	30	1645	147024	14378.9
20	30	1645	147024	14378.9
30	40	1645	147024	14378.9
40	0	1645	147024	14378.9



40	50	1645	147024	14378.9
50	60	1645	147024	14378.9

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4930.000 mm
Center of Gravity of the Platforms	4980.000 mm
Center of Gravity of the Liquid	4930.000 mm
Center of Gravity of the Nozzles	5308.018 mm
Center of Gravity of the Added Weights (Operating)	4460.556 mm
Center of Gravity of the Added Weights (Empty)	4460.556 mm
Center of Gravity of Bare Shell New and Cold	4930.000 mm
Center of Gravity of Bare Shell Corroded	4930.000 mm
Vessel CG in the Operating Condition	4934.682 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4938.119 mm
Vessel CG in the Test Condition	4931.629 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

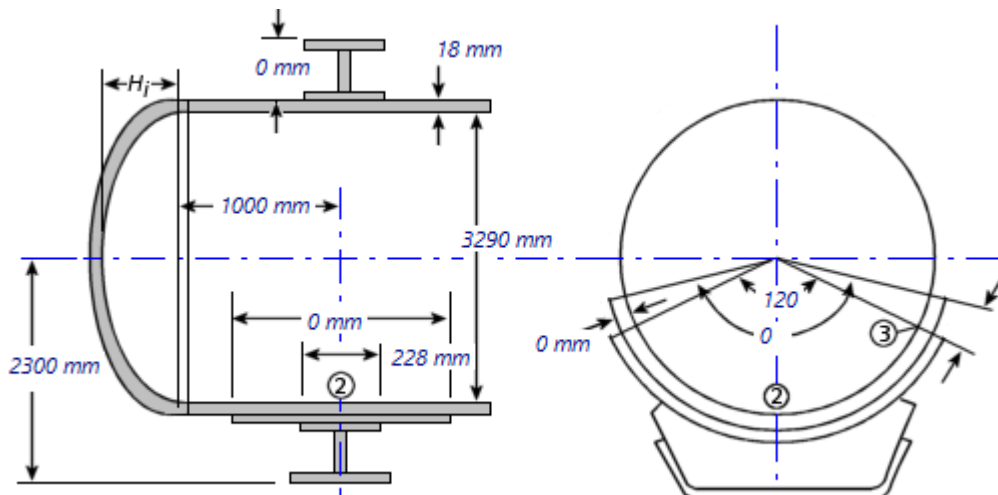
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1655.5001 \cdot 15.0}, 2 \cdot 1000.0)$$

$$= 473.8300 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1655.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	9860.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	833.00	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		656164.06	N
Pressure used in Analysis	P	1.010	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	49.07	138.00
Long. Stress at Bottom of Midspan	62.37	138.00
Long. Stress at Top of Saddles	60.44	138.00
Long. Stress at Bottom of Saddles	53.10	138.00
Tangential Shear in Shell	22.16	110.40



Circ. Stress at Horn of Saddle	149.20	172.50
Circ. Compressive Stress in Shell	7.02	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (73965.4/2 + 0) \cdot 2300.0/2897.7212$$

$$= 88062.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(29390.57, 0) \cdot 2300.0/7860.0$$

$$= 8600.3 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(100652.52, 0) \cdot 2300.0/7860.0$$

$$= 29453.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (100653/2 + 0) \cdot 2300.0/2897.7212$$

$$= 119836.0 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 536328 + \max(8600, 88063, 29453, 119836)$$

$$= 656164.1 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169 (4.324 + \pi/4 (3.326 \cdot 1.2)^2)$$

$$= 29392.092 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	663943.12	N
Transverse Shear Load	50326.26	N
Longitudinal Shear Load	100652.52	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3290.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 1655.5)$$

$$= 2897.721 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0215	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -656164 \cdot 1000.0 [1 - (1 - 1000.0/9860.0 + (1655.5^2 - 833.0^2) / (2 \cdot 1000.0 \cdot 9860.0)) / (1 + (4 \cdot 833.0)/(3 \cdot 9860.0))]$$

$$= -65056956.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 656164 \cdot 9860/4 ((1 + 2(1656^2 - 833^2)/(9860^2)) / (1 + (4 \cdot 833)/(3 \cdot 9860)) - 4 \cdot 1000/9860)$$

$$= 859089024.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$



$$= 1.01 \cdot 1655.5 / (2 \cdot 15.0) - 0.85909E+09 / (\pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 49.07 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 1655.5 / (2 \cdot 15.0) + 0.85909E+09 / (\pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 62.37 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 1655.5 / (2 \cdot 15.0) - 65056956.0 / (0.1066 \cdot \pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 60.44 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 1655.5 / (2 \cdot 15.0) + 65056956.0 / (0.1923 \cdot \pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 53.10 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 656164(9860.0 - 2 \cdot 1000.0) / (9860.0 + (4 \cdot 833.0 / 3))$$

$$= 470112.7 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 470112.69 / (1655.5001 \cdot 15.0)$$

$$= 22.16 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1655.5 \cdot 15.0)}$$

$$= 122.915 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 656164 \cdot 0.1 / (15.0(228.0 + 122.91 + 122.91))$$

$$= -7.02 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -656164 / (4 \cdot 15.0(228.0 + 122.915 + 122.915)) -$$

$$12 \cdot 0.021 \cdot 656164 \cdot 1655.5 / (9860.0 \cdot 15.0^2)$$

$$= -149.20 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1655.5 \cdot 15.0)}, 2 \cdot 1000.0)$$

$$= 473.83 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 7860.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 7860.0(85.0 - 21.1)$$

$$= 6.028 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2922.0002	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	619.0	mm
Friction Coefficient	mu	0.000	

**Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)**

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1648.0 \cdot 15.0)} + 280.0)15.0^3)/12$$

$$= 147732.734 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 147732.734 + 7879.087(250.671 - 7.5)^2$$

$$= 466053568.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	525.3	15.0	7.5	7879.1	59093.2	0.466E+09
Wearplate	280.0	10.0	20.0	2800.0	56000.0	0.149E+09
Web	16.0	609.0	329.5	9744.0	3210648.5	0.362E+09
BasePlate	254.0	18.0	643.0	4572.0	2939796.2	0.704E+09
Totals	...	...	...	24995.1	6265538.0	0.168E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6265538/24995.09$$

$$= 250.671 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 656164$$

$$= 133543.6250 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.8029 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1648.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 937.115 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 125196488.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 18.6675 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(656164 + 7779)254.0 / (4 \cdot 2922.0 \cdot 161.332))}$$

$$= 16.381 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1645.0 + 18.0 + 10.0)$$

$$= 2897.721 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$



$$= 2897.7209 / (4 - 1)$$

$$= 965.907 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 965.907 \cdot 254.0 / 2$$

$$= 122670.188 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 663943 / (2922.0 \cdot 254.0)$$

$$= 0.895 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 122670.2 \cdot 0.89$$

$$= 109737.727 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2756.0 + 7727.256$$

$$= 10483.257 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 109737.7 / 10483.2568$$

$$= 10.469 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 10483.257$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(12840037 / 10483.257)}$$

$$= 34.997 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300.0 - \cos(120.0/2) (1645.0 + 18.0 + 10.0) - 18.0$$

$$= 1445.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 242.0)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1445.5 / 34.997$$



$$= 41.303$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 100652.5 / (2 \cdot 2922.0) \cdot 965.907 \cdot 1445.5 / 2$$

$$= 12028567.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (41.3031 < 127.7103) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (41.3)^2 / (2 \cdot 127.71^2)) 242 /$$

$$(5/3 + 3 \cdot (41.3) / (8 \cdot 127.71) - (41.3)^3 / (8 \cdot 127.71^3))$$

$$= 128.6 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.47 / 128.57 + (12028567 \cdot 114.0 / 12840049) / 161.33$$

$$= 0.743$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	965.9	16.0	0.0	15454.5	0.0	0.330E+06
Totals	...	...	...	18210.5	...	0.132E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 18210.512$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 2300.0 - \sqrt{ (1673.0 + 10.0)^2 - (965.907/2)^2 } - 18.0$$

$$= 680.224 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 13165296 / 18210.512 }$$

$$= 26.888 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 680.224 / 26.888$$

$$= 25.299$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 100652.516 / (2 \cdot 2922.0)$$

$$= 17.223 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 17.223 \cdot 965.907 \cdot 680.224$$

$$= 11320824.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 11320823 / 115485.156$$

$$= 97.997 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (25.2987 < 127.7103) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



$$= (1 - (25.3)^2 / (2 \cdot 127.71^2))^{242} / (5/3 + 3 \cdot (25.3) / (8 \cdot 127.71) - (25.3^3) / (8 \cdot 127.71^3))$$

$$= 136.4 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 12.05/136.35 + (11320823 \cdot 114.0/13165296) / 161.33$$

$$= 0.696$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.90 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	91.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	477.3824 mm ²
Saddle Load Q0 (Weight)	Q0	544107.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	29453.0 N
Maximum Transverse Force	Ft	50326.3 N
Maximum Longitudinal Force	Fl	100652.5 N
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

$$= 163605.6 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= \max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 163605.6 / (477.382 \cdot 4)$$

$$= 85.7 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 50326.3 / (477.382 \cdot 4)$$

$$= 26.4 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 100652.52 / (99.9 \cdot 4.0)$$

$$= 251.9046 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 50326.26 + 0.0$$

$$= 115797328.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0.11580E+09/544107$$

$$= 212.73 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 251.9046, 0.0]$$



= 251.9046 mm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, $t_g = 10.0$, $c = 3.0$ mm, $E^* = 0.85$

Thickness Ratio = $t_r = E^*/(t_g - c) = 0.85$, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$

= $\min(228.0 + 1.56 \cdot \sqrt{1655.5001 \cdot 15.0}, 2 \cdot 1000.0)$

= 473.8300 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1655.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	9860.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	833.00	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		657408.81	N
Pressure used in Analysis	P	1.010	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	49.05	138.00
Long. Stress at Bottom of Midspan	62.38	138.00
Long. Stress at Top of Saddles	60.45	138.00
Long. Stress at Bottom of Saddles	53.09	138.00
Tangential Shear in Shell	22.21	110.40
Circ. Stress at Horn of Saddle	149.48	172.50
Circ. Compressive Stress in Shell	7.03	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$

= $3.0(73965.4/2 + 0) \cdot 2300.0/2897.7212$

= 88062.6 N

Longitudinal Saddle Reaction Force [Fwl]:



$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(29390.57, 0) \cdot 2300.0/7860.0$$

$$= 8600.3 \text{ N}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(100652.52, 0) \cdot 2300.0/7860.0$$

$$= 29453.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(100653/2 + 0) \cdot 2300.0/2897.7212$$

$$= 119836.0 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 537573 + \max(8600, 88063, 29453, 119836)$$

$$= 657408.8 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4)(OD \cdot \text{WindDiaMult})^2)$$

$$= 1.1 \cdot 1587.169(4.324 + \pi/4(3.326 \cdot 1.2)^2)$$

$$= 29392.092 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	665187.88	N
Transverse Shear Load	50326.26	N
Longitudinal Shear Load	100652.52	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3290.0/2 + 18.0 + 10.0)\sin(120.0/2), 2 \cdot 1655.5)$$

$$= 2897.721 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0215	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -657409 \cdot 1000.0 [1 - (1 - 1000.0/9860.0 + (1655.5^2 - 833.0^2)/(2 \cdot 1000.0 \cdot 9860.0))/(1 + (4 \cdot 833.0)/(3 \cdot 9860.0))]$$

$$= -65180368.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 657409 \cdot 9860/4((1 + 2(1656^2 - 833^2)/(9860^2))/(1 + (4 \cdot 833)/(3 \cdot 9860)) - 4 \cdot 1000/9860)$$

$$= 860718720.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 1.01 \cdot 1655.5/(2 \cdot 15.0) - 0.86072E+09/(\pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 49.05 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 1655.5/(2 \cdot 15.0) + 0.86072E+09/(\pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 62.38 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 1.01 \cdot 1655.5/(2 \cdot 15.0) - 65180368.0/(0.1066 \cdot \pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 60.45 \text{ MPa}$$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.01 \cdot 1655.5 / (2 \cdot 15.0) + 65180368.0 / (0.1923 \cdot \pi \cdot 1655.5^2 \cdot 15.0)$$

$$= 53.09 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 657409 (9860.0 - 2 \cdot 1000.0) / (9860.0 + (4 \cdot 833.0 / 3))$$

$$= 471004.5 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 471004.47 / (1655.5001 \cdot 15.0)$$

$$= 22.21 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{1655.5 \cdot 15.0}$$

$$= 122.915 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 657409 \cdot 0.1 / (15.0(228.0 + 122.91 + 122.91))$$

$$= -7.03 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -657409 / (4 \cdot 15.0(228.0 + 122.915 + 122.915)) - 12 \cdot 0.021 \cdot 657409 \cdot 1655.5 / (9860.0 \cdot 15.0^2)$$

$$= -149.48 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1655.5 \cdot 15.0}, 2 \cdot 1000.0)$$

$$= 473.83 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2922.0002	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	MPa
Height of Web at Center	Hw,c	619.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1648.0 \cdot 15.0}) + 280.0)^3 / 12$$

$$= 147732.734 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 147732.734 + 7879.087(250.671 - 7.5)^2$$

$$= 466053568.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	525.3	15.0	7.5	7879.1	59093.2	0.466E+09
Wearplate	280.0	10.0	20.0	2800.0	56000.0	0.149E+09
Web	16.0	609.0	329.5	9744.0	3210648.5	0.362E+09



BasePlate	254.0	18.0	643.0	4572.0	2939796.2	0.704E+09
Totals	...	...	...	24995.1	6265538.0	0.168E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6265538/24995.09$$

$$= 250.671 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 657409$$

$$= 133796.9531 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.8177 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1648.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 937.115 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 125433976.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 18.7029 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(657409 + 7779)254.0 / (4 \cdot 2922.0 \cdot 161.332)}$$

$$= 16.396 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1645.0 + 18.0 + 10.0)$$

$$= 2897.721 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2897.7209 / (4 - 1)$$

$$= 965.907 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 965.907 \cdot 254.0/2$$

$$= 122670.188 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 665188 / (2922.0 \cdot 254.0)$$

$$= 0.896 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 122670.2 \cdot 0.9$$

$$= 109943.469 \text{ N}$$

Area of the Rib and Web [Ar]:



$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 2756.0 + 7727.256 \\
 &= 10483.257 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/A_r \\
 &= 109943.5/10483.2568 \\
 &= 10.488 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 228.0 - 16.0 \\
 &= 212.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/10483.257 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 228.0/2 \\
 &= 114.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(12840037/10483.257)} \\
 &= 34.997 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2300.0 - \cos(120.0/2) (1645.0 + 18.0 + 10.0) - 18.0 \\
 &= 1445.500 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 199955/242.0)} \\
 &= 127.710
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1445.5/34.997 \\
 &= 41.303
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 100652.5 / (2 \cdot 2922.0) \cdot 965.907 \cdot 1445.5/2 \\
 &= 12028567.000 \text{ N-mm}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($41.3031 < 127.7103$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (41.3)^2 / (2 \cdot 127.71^2)) 242 / \\
 &\quad (5/3 + 3 \cdot (41.3) / (8 \cdot 127.71) - (41.3)^3 / (8 \cdot 127.71^3)) \\
 &= 128.6 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 10.49/128.57 + (12028567 \cdot 114.0/12840049) / 161.33 \\
 &= 0.743
 \end{aligned}$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	965.9	16.0	0.0	15454.5	0.0	0.330E+06
Totals	...	...	...	18210.5	...	0.132E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/18210.512$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2300.0 - \sqrt{(1673.0 + 10.0)^2 - (965.907/2)^2} - 18.0$$

$$= 680.224 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13165296/18210.512)}$$

$$= 26.888 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 680.224/26.888$$

$$= 25.299$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 100652.516/(2 \cdot 2922.0)$$

$$= 17.223 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 17.223 \cdot 965.907 \cdot 680.224$$

$$= 11320824.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 11320823/115485.156$$

$$= 97.997 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (25.2987 < 127.7103) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (25.3)^2 / (2 \cdot 127.71^2)) 242 / (5/3 + 3 \cdot (25.3) / (8 \cdot 127.71) - (25.3^3) / (8 \cdot 127.71^3))$$

$$= 136.4 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 12.07/136.35 + (11320823 \cdot 114.0/13165296) / 161.33$$

$$= 0.696$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.90 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	91.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm



Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	477.3824 mm ²
Saddle Load QO (Weight)	QO	545351.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	29453.0 N
Maximum Transverse Force	Ft	50326.3 N
Maximum Longitudinal Force	Fl	100652.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= Fl / (Bolt Area • Number of Bolts)

= 100652.5 / (477.382 • 4)

= 52.7 MPa. Must be less than 99.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 50326.3 / (477.382 • 4)

= 26.4 MPa. Must be less than 99.9 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)

= 100652.52 / (99.9 • 4.0)

= 251.9046 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 2300.0 • 50326.26 + 0.0

= 115797328.00 N-mm

Eccentricity (e):

= Rmom / QO

= 0.11580E+09/545352

= 212.25 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 251.9046, 0.0]

= 251.9046 mm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 10.0, c = 3.0 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.85, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B

-29 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-37 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

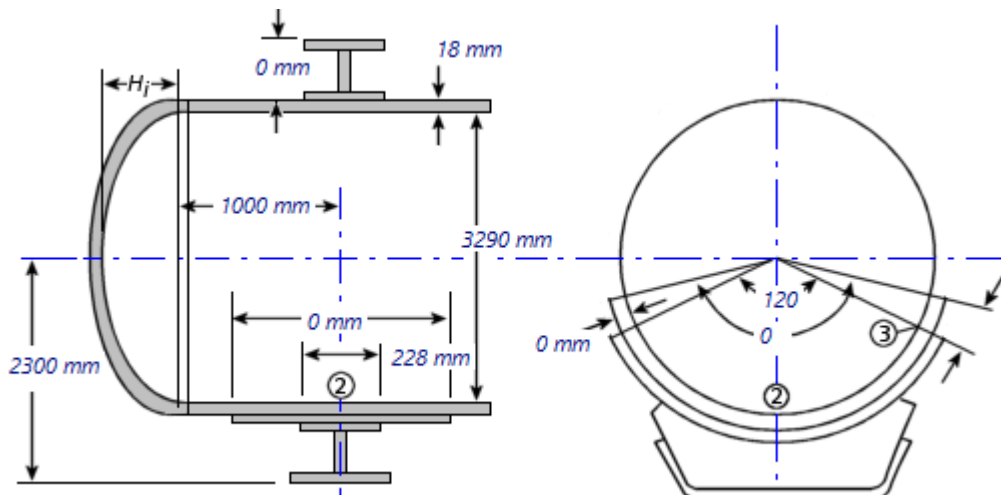
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1654.0 \cdot 18.0}, 2 \cdot 1000.0)$$

$$= 497.1712 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	R_m	1654.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	9860.00	mm
Distance from Saddle to Vessel tangent	a_1 or a	1000.00	mm
Saddle Width	b_1 or b	228.00	mm
Saddle Bearing Angle	δ or θ	120.00	degrees
Inside or Mean Depth of Head	H_i or h_m	831.50	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	μ	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	622026.38	N
Pressure used in Analysis	P	1.363 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	57.36	234.00
Long. Stress at Bottom of Midspan	67.89	234.00
Long. Stress at Top of Saddles	66.36	234.00
Long. Stress at Bottom of Saddles	60.55	234.00
Tangential Shear in Shell	17.53	138.58



Circ. Stress at Horn of Saddle	100.49	350.99
Circ. Compressive Stress in Shell	5.28	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (24408.6/2 + 0) \cdot 2300.0/2897.7212$$

$$= 29060.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(9698.89, 0) \cdot 2300.0/7860.0$$

$$= 2838.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 592966 + \max(2838, 29061, 0, 0)$$

$$= 622026.4 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169 (4.324 + \pi/4 (3.326 \cdot 1.2)^2)$$

$$= 29392.092 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	629805.44	N
Transverse Shear Load	12204.29	N
Longitudinal Shear Load	9698.89	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3290.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 1654.0)$$

$$= 2897.721 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0215	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -622026 \cdot 1000.0 [1 - (1 - 1000.0/9860.0 + (1654.0^2 - 831.5^2) / (2 \cdot 1000.0 \cdot 9860.0)) / (1 + (4 \cdot 831.5)/(3 \cdot 9860.0))]$$

$$= -61639984.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 622026 \cdot 9860/4 ((1 + 2(1654^2 - 832^2)/(9860^2)) / (1 + (4 \cdot 832)/(3 \cdot 9860)) - 4 \cdot 1000/9860)$$

$$= 814585984.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 1.363 \cdot 1654.0/(2 \cdot 18.0) - 0.81459E+09/(\pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 57.36 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 1.363 \cdot 1654.0/(2 \cdot 18.0) + 0.81459E+09/(\pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 67.89 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$



$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) - 61639984.0 / (0.1066 \cdot \pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 66.36 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) + 61639984.0 / (0.1923 \cdot \pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 60.55 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 622026(9860.0 - 2 \cdot 1000.0) / (9860.0 + (4 \cdot 831.5 / 3))$$

$$= 445735.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 445735.78 / (1654.0 \cdot 18.0)$$

$$= 17.53 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1654.0 \cdot 18.0)}$$

$$= 134.586 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 622026 \cdot 0.1 / (18.0(228.0 + 134.59 + 134.59))$$

$$= -5.28 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -622026 / (4 \cdot 18.0(228.0 + 134.586 + 134.586)) -$$

$$12 \cdot 0.022 \cdot 622026 \cdot 1654.0 / (9860.0 \cdot 18.0^2)$$

$$= -100.49 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1654.0 \cdot 18.0)}, 2 \cdot 1000.0)$$

$$= 497.17 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2922.0002	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	619.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1645.0 \cdot 18.0)} + 280.0)^3) / 12$$

$$= 266540.594 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 266540.594 + 9871.883(235.166 - 9.0)^2$$

$$= 505224608.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	548.4	18.0	9.0	9871.9	88846.9	0.505E+09
Wearplate	280.0	10.0	23.0	2800.0	64400.0	0.126E+09
Web	16.0	609.0	332.5	9744.0	3239880.8	0.393E+09



BasePlate	254.0	18.0	646.0	4572.0	2953512.2	0.772E+09
Totals	...	...	...	26987.9	6346640.0	0.180E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6346640/26987.885$$

$$= 235.166 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 622026$$

$$= 126595.8594 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 7.3970 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1645.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 939.596 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 118997192.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.5715 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(622026 + 7779) \cdot 254.0 / (4 \cdot 2922.0 \cdot 173.33)}$$

$$= 15.392 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1645.0 + 18.0 + 10.0)$$

$$= 2897.721 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2897.7209 / (4 - 1)$$

$$= 965.907 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 965.907 \cdot 254.0/2$$

$$= 122670.188 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 629805 / (2922.0 \cdot 254.0)$$

$$= 0.849 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 122670.2 \cdot 0.85$$

$$= 104095.391 \text{ N}$$

Area of the Rib and Web [Ar]:



$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 2756.0 + 7727.256 \\
 &= 10483.257 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/A_r \\
 &= 104095.4/10483.2568 \\
 &= 9.931 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 228.0 - 16.0 \\
 &= 212.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/10483.257 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 228.0/2 \\
 &= 114.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(12840037/10483.257)} \\
 &= 34.997 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2300.0 - \cos(120.0/2) (1645.0 + 18.0 + 10.0) - 18.0 \\
 &= 1445.500 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 199955/260.0)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1445.5/34.997 \\
 &= 41.303
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 9698.9 / (2 \cdot 2922.0) \cdot 965.907 \cdot 1445.5/2 \\
 &= 1159074.250 \text{ N-mm}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($41.3031 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (41.3)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (41.3) / (8 \cdot 123.21) - (41.3^3) / (8 \cdot 123.21^3)) \\
 &= 137.3 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 9.93/137.27 + (1159074 \cdot 114.0/12840049) / 173.33 \\
 &= 0.132
 \end{aligned}$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	965.9	16.0	0.0	15454.5	0.0	0.330E+06
Totals	...	...	...	18210.5	...	0.132E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/18210.512$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2300.0 - \sqrt{(1673.0 + 10.0)^2 - (965.907/2)^2} - 18.0$$

$$= 680.224 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13165296/18210.512)}$$

$$= 26.888 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 680.224/26.888$$

$$= 25.299$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 9698.889/(2 \cdot 2922.0)$$

$$= 1.660 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.66 \cdot 965.907 \cdot 680.224$$

$$= 1090875.875 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1090876/115485.156$$

$$= 9.443 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (25.2987 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (25.3)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (25.3) / (8 \cdot 123.21) - (25.3^3) / (8 \cdot 123.21^3))$$

$$= 146.1 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11.43/146.06 + (1090876 \cdot 114.0/13165296) / 173.33$$

$$= 0.133$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	91.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric



BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	477.3824 mm ²
Saddle Load QO (Weight)	QO	600744.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	2838.1 N
Maximum Transverse Force	Ft	12204.3 N
Maximum Longitudinal Force	F1	9698.9 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9698.9 / (477.382 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 12204.3 / (477.382 \cdot 4)$$

$$= 6.4 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 9698.89 / (68.4 \cdot 4.0)$$

$$= 35.4528 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 12204.29 + 0.0$$

$$= 28081260.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 28081260 / 600745$$

$$= 46.73 \text{ mm} < Bplen / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 35.4528, 0.0]$$

$$= 35.4528 \text{ mm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{ 1654.0 \cdot 18.0 }, 2 \cdot 1000.0)$$

$$= 497.1712 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1654.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	9860.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1000.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees



Inside or Mean Depth of Head	Hi or hm	831.50	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		623271.00	N
Pressure used in Analysis	P	1.363	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	57.35	234.00
Long. Stress at Bottom of Midspan	67.90	234.00
Long. Stress at Top of Saddles	66.37	234.00
Long. Stress at Bottom of Saddles	60.55	234.00
Tangential Shear in Shell	17.56	138.58
Circ. Stress at Horn of Saddle	100.69	350.99
Circ. Compressive Stress in Shell	5.30	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (24408.6/2 + 0) \cdot 2300.0/2897.7212$
 $= 29060.6 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(9698.89, 0) \cdot 2300.0/7860.0$
 $= 2838.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 594210 + \max(2838, 29061, 0, 0)$
 $= 623271.0 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (4.324 + \pi/4 (3.326 \cdot 1.2)^2)$
 $= 29392.092 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	631050.06	N
Transverse Shear Load	12204.29	N
Longitudinal Shear Load	9698.89	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(3290.0/2 + 18.0 + 10.0) \sin(120.0/2), 2 \cdot 1654.0)$
 $= 2897.721 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0215	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.



Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -623271 \cdot 1000.0 \left[1 - \left(1 - \frac{1000.0}{9860.0} + \frac{(1654.0^2 - 831.5^2)}{(2 \cdot 1000.0 \cdot 9860.0)} \right) / \left(1 + \frac{(4 \cdot 831.5)}{(3 \cdot 9860.0)} \right) \right]$$

$$= -61763320.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 623271 \cdot 9860 / 4 \left(\frac{(1 + 2(1654^2 - 832^2)/(9860^2))}{(1 + (4 \cdot 832)/(3 \cdot 9860))} - 4 \cdot 1000 / 9860 \right)$$

$$= 816216000.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) - 0.81622E+09 / (\pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 57.35 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) + 0.81622E+09 / (\pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 67.90 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) - 61763320.0 / (0.1066 \cdot \pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 66.37 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1.363 \cdot 1654.0 / (2 \cdot 18.0) + 61763320.0 / (0.1923 \cdot \pi \cdot 1654.0^2 \cdot 18.0)$$

$$= 60.55 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 623271(9860.0 - 2 \cdot 1000.0) / (9860.0 + (4 \cdot 831.5 / 3))$$

$$= 446627.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 446627.62 / (1654.0 \cdot 18.0)$$

$$= 17.56 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1654.0 \cdot 18.0)}$$

$$= 134.586 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 623271 \cdot 0.1 / (18.0(228.0 + 134.59 + 134.59))$$

$$= -5.30 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -623271 / (4 \cdot 18.0(228.0 + 134.586 + 134.586)) - 12 \cdot 0.022 \cdot 623271 \cdot 1654.0 / (9860.0 \cdot 18.0^2)$$

$$= -100.69 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1654.0 \cdot 18.0)}, 2 \cdot 1000.0)$$

$$= 497.17 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2922.0002	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	254.0000	mm



Number of Ribs (inc. outside ribs)	Nribs	4
Rib Thickness	Ribtk	13.0000 mm
Web Thickness	Webtk	16.0000 mm
Web Location	Webloc	Center
Saddle Yield Stress	Sy	260.0 MPa
Height of Web at Center	Hw,c	619.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1645.0 \cdot 18.0)} + 280.0) 18.0^3) / 12$$

$$= 266540.594 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 266540.594 + 9871.883 (235.166 - 9.0)^2$$

$$= 505224608.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	548.4	18.0	9.0	9871.9	88846.9	0.505E+09
Wearplate	280.0	10.0	23.0	2800.0	64400.0	0.126E+09
Web	16.0	609.0	332.5	9744.0	3239880.8	0.393E+09
BasePlate	254.0	18.0	646.0	4572.0	2953512.2	0.772E+09
Totals	...	...	...	26987.9	6346640.0	0.180E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6346640 / 26987.885$$

$$= 235.166 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 623271$$

$$= 126849.1562 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.4118 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1645.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 939.596 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 119235296.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.6026 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(623271 + 7779) 254.0 / (4 \cdot 2922.0 \cdot 173.33)}$$

$$= 15.407 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:



Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1645.0 + 18.0 + 10.0)$$

$$= 2897.721 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2897.7209 / (4 - 1)$$

$$= 965.907 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 965.907 \cdot 254.0 / 2$$

$$= 122670.188 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 631050 / (2922.0 \cdot 254.0)$$

$$= 0.850 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 122670.2 \cdot 0.85$$

$$= 104301.102 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2756.0 + 7727.256$$

$$= 10483.257 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 104301.1 / 10483.2568$$

$$= 9.950 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 10483.257$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(12840037 / 10483.257)}$$

$$= 34.997 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300.0 - \cos(120.0/2) (1645.0 + 18.0 + 10.0) - 18.0$$

$$= 1445.500 \text{ mm}$$

Intermediate Term [Cc]:



$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1445.5 / 34.997$$

$$= 41.303$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 9698.9 / (2 \cdot 2922.0) \cdot 965.907 \cdot 1445.5 / 2$$

$$= 1159074.250 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (41.3031 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (41.3)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (41.3) / (8 \cdot 123.21) - (41.3^3) / (8 \cdot 123.21^3))$$

$$= 137.3 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.95 / 137.27 + (1159074 \cdot 114.0 / 12840049) / 173.33$$

$$= 0.132$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	965.9	16.0	0.0	15454.5	0.0	0.330E+06
Totals	...	...	...	18210.5	...	0.132E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 18210.512$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{\text{pdthk}})^2 - (\text{Pitch}/2)^2} - B_{\text{pthk}}$$

$$= 2300.0 - \sqrt{(1673.0 + 10.0)^2 - (965.907/2)^2} - 18.0$$

$$= 680.224 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13165296 / 18210.512)}$$

$$= 26.888 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 680.224 / 26.888$$

$$= 25.299$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 9698.889 / (2 \cdot 2922.0)$$

$$= 1.660 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.66 \cdot 965.907 \cdot 680.224$$

$$= 1090875.875 \text{ N-mm}$$



Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1090876 / 115485.156$$

$$= 9.443 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($25.2987 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (25.3)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (25.3) / (8 \cdot 123.21) - (25.3^3) / (8 \cdot 123.21^3))$$

$$= 146.1 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 11.45 / 146.06 + (1090876 \cdot 114.0 / 13165296) / 173.33$$

$$= 0.133$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	91.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	477.3824 mm ²
Saddle Load QO (Weight)	QO	601989.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	2838.1 N
Maximum Transverse Force	Ft	12204.3 N
Maximum Longitudinal Force	Fl	9698.9 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [$\tau_{b,l}$]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9698.9 / (477.382 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [$\tau_{b,t}$]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 12204.3 / (477.382 \cdot 4)$$

$$= 6.4 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 9698.89 / (68.4 \cdot 4.0)$$

$$= 35.4528 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 12204.29 + 0.0$$

$$= 28081260.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 28081260 / 601989$$

$$= 46.63 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User

FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 75 of 181

Saddle Calcs: Test Case: Step: 18 2:01pm Sep 17,2026



HEXAGON

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 35.4528, 0.0]
= 35.4528 mm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

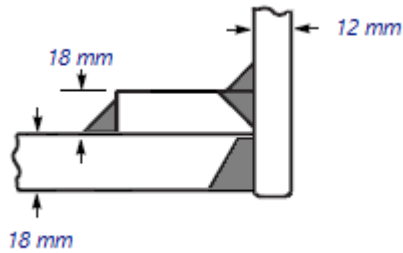
**Input, Nozzle Desc: M1-24''****From: 10**

Pressure for Reinforcement Calculations	P	1.0457	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3290.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.05 \cdot 3296.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.05)$
 $= 11.2280 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.05 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 1.05)$
 $= 2.3027 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001	mm
Parallel to Vessel Wall, opening length	d	591.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 138.0/138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 138.0/138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, Sp/Sv)$$

$$= \min(1, 138.0/138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

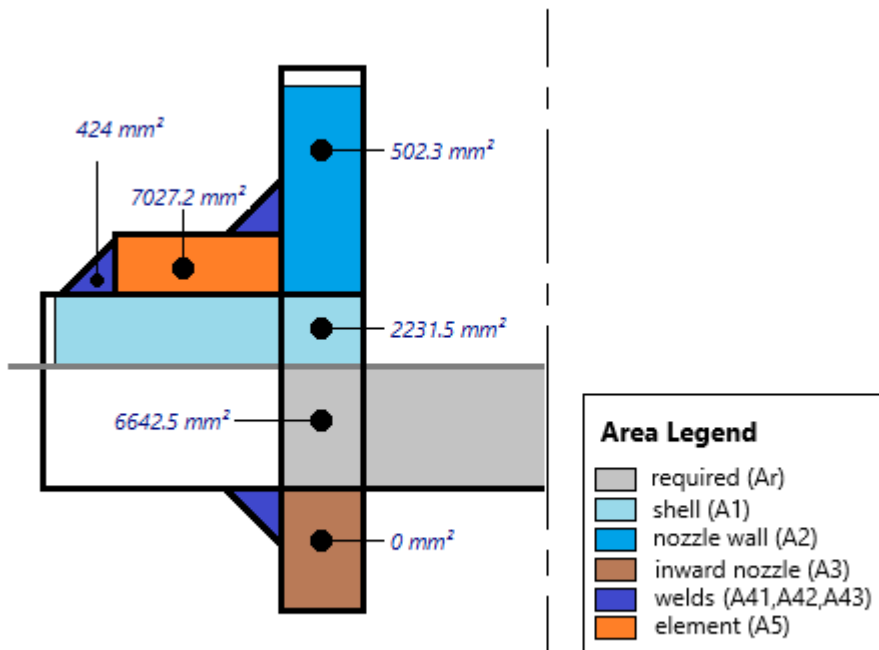


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6642.510	NA	NA
Area in Shell	A1	2231.491	NA	NA
Area in Nozzle Wall	A2	502.300	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	424.000	NA	NA
Area in Element	A5	7027.200	NA	NA
TOTAL AREA AVAILABLE	Atot	10184.991	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	803.1954	18.0000 mm
Based on given Pad Diameter:	1000.0000	8.9260 mm
Based on Shell or Nozzle Thickness:	899.9932	12.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (591.6 \cdot 11.228 \cdot 1.0 + 2 \cdot 9.0 \cdot 11.228 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 6642.510 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 591.6 (1.0 \cdot 15.0 - 1.0 \cdot 11.228) - 2 \cdot 9.0$$

$$(1.0 \cdot 15.0 - 1.0 \cdot 11.228) \cdot (1 - 1.0)$$

$$= 2231.491 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 37.5) \cdot (9.0 - 2.3) \cdot 1.0$$

$$= 502.300 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$$

$$= 424.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (1000.0 - 609.6) \cdot 18.0 \cdot 1.0$$

$$= 7027.200 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.3027 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.3027 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 11.228$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.749$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1.05/1.96 = 0.533

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	8.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (6642.5098 - 2231.491 + 2 * 9.0 * 1.0 *
(1.0 * 15.0 - 11.228))138)
= 618026.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²*fr2)*Sv
= (502.3004 + 7027.2002 + 424.0 - 0.0 * 1.0) * 138
= 1097470.00 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (502.3004 + 0.0 + 100.0 + (270.0)) * 138
= 120365.03 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (502.3004 + 0.0 + 424.0 + 7027.2002 + (270.0)) * 138
= 1134726.12 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 609.6 * 10.0 * 0.49 * 138
= 647434. N

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 1000.0 * 18.0 * 0.49 * 138
= 1911713. N

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 300.3) * (12.0 - 3.0) * 0.7 * 138
= 820125. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 609.6 * 18.0 * 0.74 * 138
= 1759962. N



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 1466635. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (1911713 + 820125) = 2731838 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (647434 + 1759962 + 1466635 + 0) = 3874031 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (1911713 + 1466635 + 0) = 3378349 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2731838 N	, must exceed W = 618026 N	or W1 = 1097470 N
Path 2-2 = 3874031 N	, must exceed W = 618026 N	or W2 = 120365 N
Path 3-3 = 3378348 N	, must exceed W = 618026 N	or W3 = 1134726 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 15.6529 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 333.6530 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K2A-2"

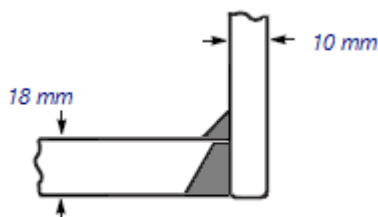
From: 20

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
= 12.4285 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.04 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
= 0.2222 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+t	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2222	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.4285	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.4285	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.42, max(15.4285, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.2222, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.222, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.04 / 1.96 = 0.529$

Weld Size Calculations, Description: K2A-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
---	------	--------	----

**Results Per UW-16.1:**

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1962 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 40.40 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K2B-2''

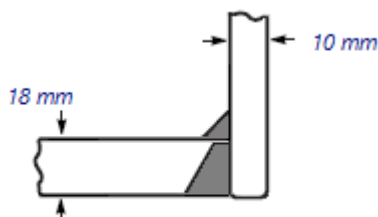
From: 20

Pressure for Reinforcement Calculations	P	1.0618	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation

0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)= $1.06 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.06)$

= 12.7392 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)= $1.06 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.06)$

= 0.2277 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 80.8000 mm

Parallel to Vessel Wall Rn+tn+t 40.4000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2277 mm

Wall Thickness per UG16(b), tr16b = 4.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 15.7392 mm

Wall Thickness tb1 = max(trb1, tr16b) = 15.7392 mm

Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm

Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[6.42, max(15.7392, 4.5)]

= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.2277, 6.42)

= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.228, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.06 / 1.96 = 0.542$ **Weld Size Calculations, Description: K2B-2''**

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1962 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

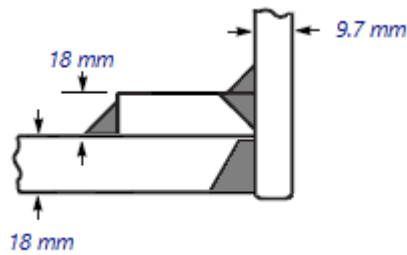
**Input, Nozzle Desc: N3-4''****From: 20**

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
 $= 12.4285 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.04 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
 $= 0.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618 mm
Parallel to Vessel Wall, opening length	d	100.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.8364 mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.855, 1.0)$$

$$= 0.855$$

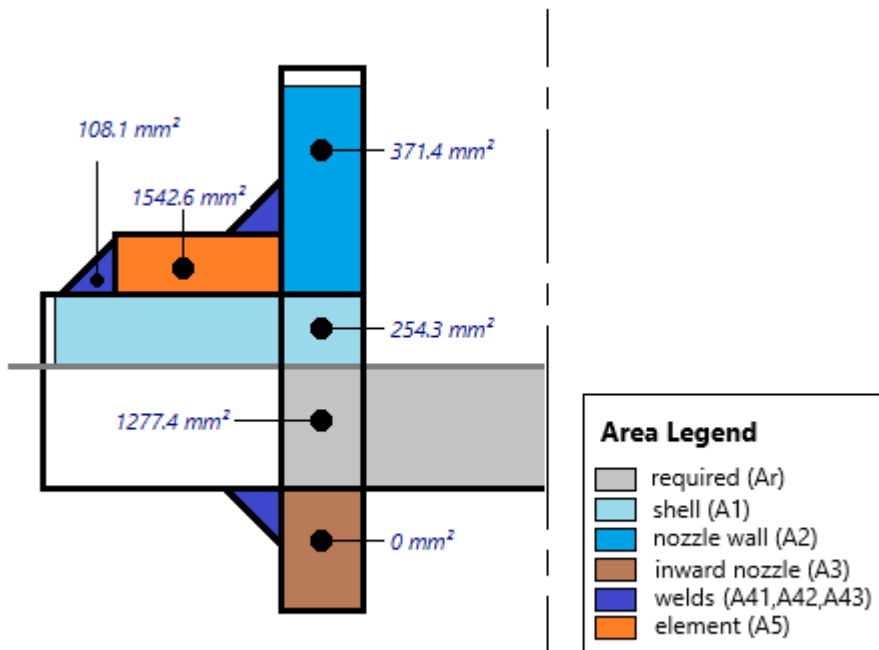


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1277.438	NA	NA
Area in Shell	A1	254.267	NA	NA
Area in Nozzle Wall	A2	371.423	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	108.082	NA	NA
Area in Element	A5	1542.600	NA	NA
TOTAL AREA AVAILABLE	Atot	2276.372	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	144.5036	18.0000 mm
Based on given Pad Diameter:	200.0000	6.3438 mm
Based on Shell or Nozzle Thickness:	170.1491	9.7345 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (100.8309 \cdot 12.4285 \cdot 1.0 + 2 \cdot 6.7345 \cdot 12.4285 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1277.438 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 100.831(1.0 \cdot 15.0 - 1.0 \cdot 12.429) - 2 \cdot 6.735$
 $(1.0 \cdot 15.0 - 1.0 \cdot 12.4285) \cdot (1 - 0.855)$
 $= 254.267 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 34.84) \cdot (6.73 - 0.5) \cdot 0.8551$$

$$= 371.423 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100.0) \cdot 0.86 + (0.0) \cdot 0.86 + 22.5746^2 \cdot 1.0$$

$$= 108.082 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot \text{fr4}$$

$$= (200.0 - 114.3) \cdot 18.0 \cdot 1.0$$

$$= 1542.600 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5000 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 15.4285 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 15.4285 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.258, \max(15.4285, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.4 MPa	Passed
Occasional	:	3.6,	Allowable	:	156.9 MPa	Passed
Shear	:	19.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 12.429, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.829$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.5, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.04/5.10 = 0.203$

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$7.5000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (1277.4377 - 254.267 + 2 \cdot 6.7345 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 12.4285)) \cdot 138)$
 $= 145269.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (371.4234 + 1542.6 + 108.0818 - 0.0 \cdot 0.86) \cdot 138$
 $= 279021.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (371.4234 + 0.0 + 85.5072 + (172.7558)) \cdot 138$
 $= 86887.78 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (371.4234 + 0.0 + 108.0818 + 1542.6 + (172.7558)) \cdot 138$
 $= 302859.59 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103800. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 200.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 297378. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118$$

$$= 93980. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 329993. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 274994. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (297378 + 93980) = 391358 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (103800 + 329993 + 274994 + 0) = 708788 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (297378 + 274994 + 0) = 572372 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 391357 N	, must exceed W = 145269 N	or W1 = 279021 N
Path 2-2 = 708787 N	, must exceed W = 145269 N	or W2 = 86887 N
Path 3-3 = 572371 N	, must exceed W = 145269 N	or W3 = 302859 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9930 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9930 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 100.83 mm

Input Echo, WRC297 Item 1, Description: N3-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3290.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	1.04	MPa
Radial Load	P	-3600.00	N
Circumferential Shear	Vc	4800.01	N
Longitudinal Shear	Vl	4800.01	N
Circumferential Moment	Mc	0.125E+07	N-mm
Longitudinal Moment	Ml	0.125E+07	N-mm
Torsional Moment	Mt	0.144E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1664.5(18.0 - 3.0)}$$

$$= 635.737 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3329.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	4.900	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.345$$

Nr/P	=	0.147
Mr/P	=	0.103
M0/P	=	0.246
N0/P	=	0.309
MrD/Mc	=	0.111
NrDL/Mc	=	0.135
M0d/Mc	=	0.300
N0DL/Mc	=	-0.136
MrD/Ml	=	0.111
NrDL/Ml	=	0.137
M0D/Ml	=	0.290
N0DL/Ml	=	-0.074

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	52	51	52

Outplane Stress Summary	40	66	73	30
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	25	25	25	25

Inplane Stress Summary	20	29	36	18
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40	66	73	30
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	14	41	48	2
-------------------------	----	----	----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	52	51	52
Inplane Stress Summary	45	56	61	45

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	45	56	62	45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9
Hoop Stress Summary	10	11	8	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-61	61	61	-61
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-63	32	103	-48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	14	14	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	79	39	103	60



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9
Hoop Stress Summary	11	12	8	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-62	62	62	-62
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-64	32	104	-49

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10
Shear Stress Summary	6	6	14	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	76	34	106	65

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1655.5(18.0 - 3.0))}$
 $= 634.325 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3311.000 mm
 Nozzle Outside Diameter (d) = 200.000 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 49.585 mm
 $T / t = 0.303$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.897$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.111
 Mr/P = 0.242



MØ/P = 0.070
 NØ/P = 0.129
 MrD/Mc = 0.312
 NrDL/Mc = 0.073
 MØd/Mc = 0.096
 NØDL/Mc = 0.038
 MrD/Ml = 0.209
 NrDL/Ml = 0.091
 MØD/Ml = 0.066
 NØDL/Ml = 0.105

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	113	114	113	114
Outplane Stress Summary	108	117	135	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	44	67	118	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	117	136	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	48	66	82	37

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-51	51	51	-51
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	113	114	113	114
Inplane Stress Summary	84	142	192	43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	84	142	192	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	52	51	52	51	52	51	52
Circ. Pl (SUS)	1	1	1	1	-1	-1	1	1
Circ. Q (SUS)	-13	13	21	-21	-4	4	8	-8
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0
Long. Q (SUS)	-4	4	8	-8	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	52.0	51.0	52.0	51.0	52.0	51.0	52.0
Pm+Pl (SUS)	52.0	53.0	52.0	53.0	50.0	51.0	52.0	53.0
Pm+Pl+Q (Total)	39.2	66.1	73.1	32.3	46.1	55.3	60.3	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------



Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	73.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	113	114	113	114	113	114	113	114
Circ. Pl (SUS)	0	0	4	4	-1	-1	3	3
Circ. Q (SUS)	-4	4	16	-16	-28	28	74	-74
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	-1	-1	3	3	1	1	3	3
Long. Q (SUS)	-11	11	57	-57	-9	9	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	113.0	114.0	113.0	114.0	113.0	114.0	113.0	114.0
Pm+Pl (SUS)	113.0	114.0	117.0	118.0	112.0	113.0	116.0	117.0
Pm+Pl+Q (Total)	109.1	118.1	133.0	102.0	84.0	141.0	190.0	43.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	190.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-20	-20	22	22	-20	-20	22	22
Long. Q (SUS)	-48	48	74	-74	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	25.7	26.7	26.5	26.5	26.7	27.7	26.5	26.5
Pm+Pl+Q (Total)	77.5	38.0	100.5	59.7	76.3	35.0	102.8	63.6

**Nozzle Stress Summation Comparison (MPa):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	27.66	177.00	Passed
Pm+Pl+Q (TOTAL)	102.78	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

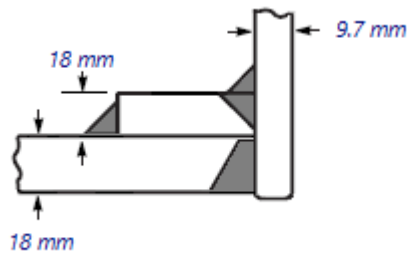
**Input, Nozzle Desc: N4-4''****From: 20**

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2950.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
 $= 12.4285 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.04 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
 $= 0.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618	mm
Parallel to Vessel Wall, opening length	d	100.8309	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.8364	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.855, 1.0)$$

$$= 0.855$$

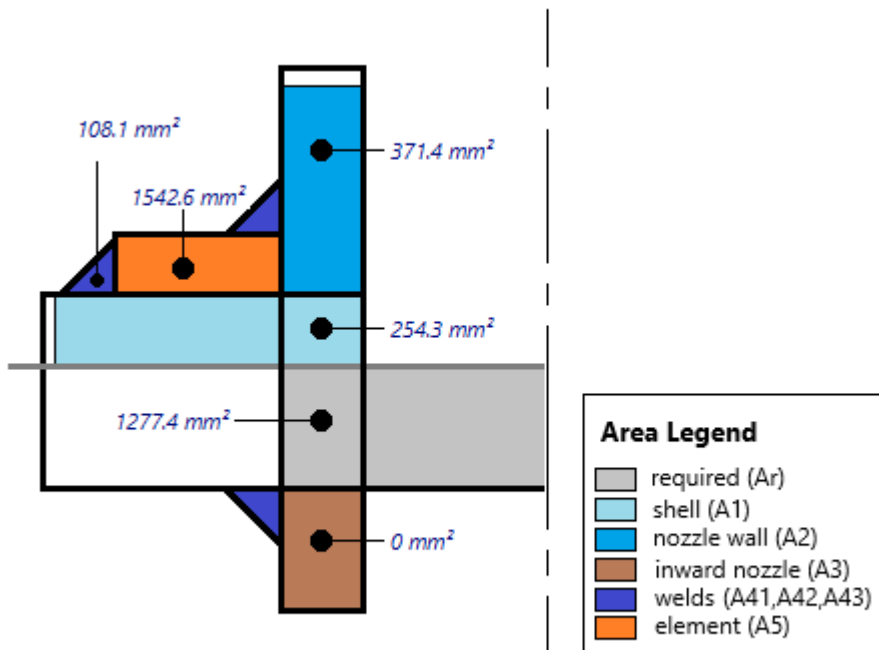


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1277.438	NA	NA
Area in Shell	A1	254.267	NA	NA
Area in Nozzle Wall	A2	371.423	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	108.082	NA	NA
Area in Element	A5	1542.600	NA	NA
TOTAL AREA AVAILABLE	Atot	2276.372	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	144.5036	18.0000 mm
Based on given Pad Diameter:	200.0000	6.3438 mm
Based on Shell or Nozzle Thickness:	170.1491	9.7345 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (100.8309 \cdot 12.4285 \cdot 1.0 + 2 \cdot 6.7345 \cdot 12.4285 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 1277.438 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 100.831 (1.0 \cdot 15.0 - 1.0 \cdot 12.429) - 2 \cdot 6.735$$

$$(1.0 \cdot 15.0 - 1.0 \cdot 12.4285) \cdot (1 - 0.855)$$

$$= 254.267 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 34.84) \cdot (6.73 - 0.5) \cdot 0.8551$$

$$= 371.423 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + Trapfr_4$$

$$= (100.0) \cdot 0.86 + (0.0) \cdot 0.86 + 22.5746^2 \cdot 1.0$$

$$= 108.082 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (200.0 - 114.3) \cdot 18.0 \cdot 1.0$$

$$= 1542.600 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5000 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 15.4285 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 15.4285 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.258, \max(15.4285, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.4 MPa	Passed
Occasional	:	3.6,	Allowable	:	156.9 MPa	Passed
Shear	:	19.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 12.429, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.829$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.5, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.04/5.10 = 0.203$

Weld Size Calculations, Description: N4-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$7.5000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (1277.4377 - 254.267 + 2 \cdot 6.7345 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 12.4285)) \cdot 138)$
 $= 145269.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (371.4234 + 1542.6 + 108.0818 - 0.0 \cdot 0.86) \cdot 138$
 $= 279021.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (371.4234 + 0.0 + 85.5072 + (172.7558)) \cdot 138$
 $= 86887.78 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (371.4234 + 0.0 + 108.0818 + 1542.6 + (172.7558)) \cdot 138$
 $= 302859.59 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103800. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 200.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 297378. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118$$

$$= 93980. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 329993. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 274994. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (297378 + 93980) = 391358 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103800 + 329993 + 274994 + 0) = 708788 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (297378 + 274994 + 0) = 572372 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 391357 N	, must exceed W = 145269 N	or W1 = 279021 N
Path 2-2 = 708787 N	, must exceed W = 145269 N	or W2 = 86887 N
Path 3-3 = 572371 N	, must exceed W = 145269 N	or W3 = 302859 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9930 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9930 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 100.83 mm

Input Echo, WRC297 Item 1, Description: N4-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3290.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	1.04	MPa
Radial Load	P	-3600.00	N
Circumferential Shear	Vc	4800.01	N
Longitudinal Shear	Vl	4800.01	N
Circumferential Moment	Mc	0.125E+07	N-mm
Longitudinal Moment	Ml	0.125E+07	N-mm
Torsional Moment	Mt	0.144E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1664.5(18.0 - 3.0)}$$

$$= 635.737 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3329.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	4.900	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.345$$

Nr/P	=	0.147
Mr/P	=	0.103
M0/P	=	0.246
N0/P	=	0.309
MrD/Mc	=	0.111
NrDL/Mc	=	0.135
M0d/Mc	=	0.300
N0DL/Mc	=	-0.136
MrD/Ml	=	0.111
NrDL/Ml	=	0.137
M0D/Ml	=	0.290
N0DL/Ml	=	-0.074

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	52	51	52

Outplane Stress Summary	40	66	73	30
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	25	25	25	25

Inplane Stress Summary	20	29	36	18
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	40	66	73	30
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	14	41	48	2
-------------------------	----	----	----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	52	51	52
Inplane Stress Summary	45	56	61	45

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	45	56	62	45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9
Hoop Stress Summary	10	11	8	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-61	61	61	-61
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-63	32	103	-48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	14	14	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	79	39	103	60



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9
Hoop Stress Summary	11	12	8	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-62	62	62	-62
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-64	32	104	-49

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10
Shear Stress Summary	6	6	14	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	76	34	106	65

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1655.5(18.0 - 3.0))}$
 $= 634.325 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3311.000 mm
 Nozzle Outside Diameter (d) = 200.000 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 49.585 mm
 T / t = 0.303
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.897$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.111
 Mr/P = 0.242



MØ/P = 0.070
 NØ/P = 0.129
 MrD/Mc = 0.312
 NrDL/Mc = 0.073
 MØd/Mc = 0.096
 NØDL/Mc = 0.038
 MrD/Ml = 0.209
 NrDL/Ml = 0.091
 MØD/Ml = 0.066
 NØDL/Ml = 0.105

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	113	114	113	114
Outplane Stress Summary	108	117	135	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	44	67	118	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	117	136	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	48	66	82	37

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-51	51	51	-51
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	113	114	113	114
Inplane Stress Summary	84	142	192	43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	84	142	192	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	52	51	52	51	52	51	52
Circ. Pl (SUS)	1	1	1	1	-1	-1	1	1
Circ. Q (SUS)	-13	13	21	-21	-4	4	8	-8
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0
Long. Q (SUS)	-4	4	8	-8	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	52.0	51.0	52.0	51.0	52.0	51.0	52.0
Pm+Pl (SUS)	52.0	53.0	52.0	53.0	50.0	51.0	52.0	53.0
Pm+Pl+Q (Total)	39.2	66.1	73.1	32.3	46.1	55.3	60.3	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------



Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	73.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	113	114	113	114	113	114	113	114
Circ. Pl (SUS)	0	0	4	4	-1	-1	3	3
Circ. Q (SUS)	-4	4	16	-16	-28	28	74	-74
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	-1	-1	3	3	1	1	3	3
Long. Q (SUS)	-11	11	57	-57	-9	9	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	113.0	114.0	113.0	114.0	113.0	114.0	113.0	114.0
Pm+Pl (SUS)	113.0	114.0	117.0	118.0	112.0	113.0	116.0	117.0
Pm+Pl+Q (Total)	109.1	118.1	133.0	102.0	84.0	141.0	190.0	43.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	190.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-20	-20	22	22	-20	-20	22	22
Long. Q (SUS)	-48	48	74	-74	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	25.7	26.7	26.5	26.5	26.7	27.7	26.5	26.5
Pm+Pl+Q (Total)	77.5	38.0	100.5	59.7	76.3	35.0	102.8	63.6



Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	27.66	177.00	Passed
Pm+Pl+Q (TOTAL)	102.78	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3-2''

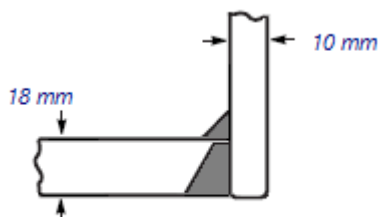
From: 30

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation

0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1) $= 1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$ $= 12.4285 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1) $= 1.04 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$ $= 0.2222 \text{ mm}$ **UG-40, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 80.8000 mm

Parallel to Vessel Wall Rn+tn+t 40.4000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2''.**This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.***UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures ta = 3.2222 mm

Wall Thickness per UG16(b), tr16b = 4.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 15.4285 mm

Wall Thickness tb1 = max(trb1, tr16b) = 15.4285 mm

Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm

Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

 $= \min[tb3, \max(tb1, tb2)]$ $= \min[6.42, \max(15.4285, 4.5)]$ $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

 $= \max(ta, tb)$ $= \max(3.2222, 6.42)$ $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.222, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.04 / 1.96 = 0.529$ **Weld Size Calculations, Description: K3-2''**

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1962 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K4-2''

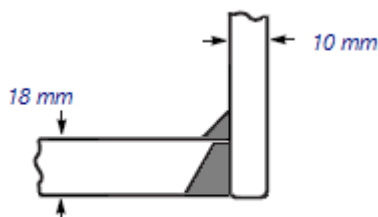
From: 30

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5450.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation

0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)= $1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$

= 12.4285 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)= $1.04 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$

= 0.2222 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 80.8000 mm

Parallel to Vessel Wall Rn+tn+t 40.4000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 17.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2222 mm

Wall Thickness per UG16(b), tr16b = 4.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 15.4285 mm

Wall Thickness tb1 = max(trb1, tr16b) = 15.4285 mm

Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm

Wall Thickness per table UG-45 tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[6.42, max(15.4285, 4.5)]

= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.2222, 6.42)

= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.222, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1.04 / 1.96 = 0.529$ **Weld Size Calculations, Description: K4-2''**

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1962 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

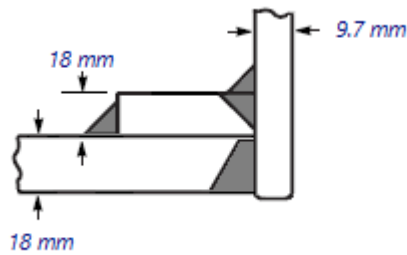
**Input, Nozzle Desc: N5-4''****From: 40**

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6550.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N5-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
 $= 12.4285 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.04 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
 $= 0.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618	mm
Parallel to Vessel Wall, opening length	d	100.8309	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.8364	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.855, 1.0)$$

$$= 0.855$$

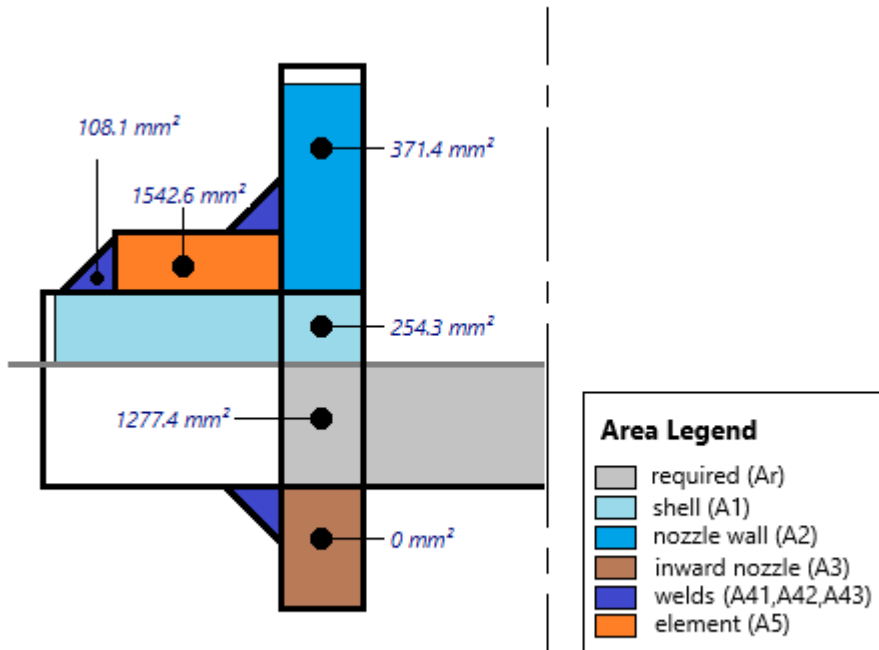


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1277.438	NA	NA
Area in Shell	A1	254.267	NA	NA
Area in Nozzle Wall	A2	371.423	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	108.082	NA	NA
Area in Element	A5	1542.600	NA	NA
TOTAL AREA AVAILABLE	Atot	2276.372	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	144.5036	18.0000 mm
Based on given Pad Diameter:	200.0000	6.3438 mm
Based on Shell or Nozzle Thickness:	170.1491	9.7345 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (100.8309 \cdot 12.4285 \cdot 1.0 + 2 \cdot 6.7345 \cdot 12.4285 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1277.438 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 100.831(1.0 \cdot 15.0 - 1.0 \cdot 12.429) - 2 \cdot 6.735$
 $(1.0 \cdot 15.0 - 1.0 \cdot 12.4285) \cdot (1 - 0.855)$
 $= 254.267 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 34.84) \cdot (6.73 - 0.5) \cdot 0.8551$$

$$= 371.423 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100.0) \cdot 0.86 + (0.0) \cdot 0.86 + 22.5746^2 \cdot 1.0$$

$$= 108.082 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot \text{fr4}$$

$$= (200.0 - 114.3) \cdot 18.0 \cdot 1.0$$

$$= 1542.600 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5000 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 15.4285 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 15.4285 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.258, \max(15.4285, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.4 MPa	Passed
Occasional	:	3.6,	Allowable	:	156.9 MPa	Passed
Shear	:	19.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 12.429, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.829, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.5, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.074, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1.04/5.10 = 0.203

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.7142 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5 * TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (1277.4377 - 254.267 + 2 * 6.7345 * 0.8551 *
(1.0 * 15.0 - 12.4285))138)
= 145269.62 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv
= (371.4234 + 1542.6 + 108.0818 - 0.0 * 0.86) * 138
= 279021.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (371.4234 + 0.0 + 85.5072 + (172.7558)) * 138
= 86887.78 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (371.4234 + 0.0 + 108.0818 + 1542.6 + (172.7558)) * 138
= 302859.59 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 103800. \text{ N}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.1416/2.0) \cdot 200.0 \cdot 14.0 \cdot 0.49 \cdot 138 \\
 &= 297378. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118 \\
 &= 93980. \text{ N}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138 \\
 &= 329993. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138 \\
 &= 274994. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (297378 + 93980) = 391358 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (103800 + 329993 + 274994 + 0) = 708788 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (297378 + 274994 + 0) = 572372 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 391357 N	, must exceed W = 145269 N	or W1 = 279021 N
Path 2-2 = 708787 N	, must exceed W = 145269 N	or W2 = 86887 N
Path 3-3 = 572371 N	, must exceed W = 145269 N	or W3 = 302859 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9930 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9930 mm

Input Echo, WRC297 Item 1, Description: N5-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3290.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.04	MPa
Radial Load	P	-3600.00	N
Circumferential Shear	Vc	4800.01	N
Longitudinal Shear	Vl	4800.01	N
Circumferential Moment	Mc	0.125E+07	N-mm
Longitudinal Moment	Ml	0.125E+07	N-mm
Torsional Moment	Mt	0.144E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1664.5(18.0 - 3.0))}$$

$$= 635.737 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3329.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	4.900	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.345$$

Nr/P	=	0.147
Mr/P	=	0.103
M0/P	=	0.246
N0/P	=	0.309
MrD/Mc	=	0.111
NrDL/Mc	=	0.135
M0d/Mc	=	0.300
N0DL/Mc	=	-0.136
MrD/Ml	=	0.111
NrDL/Ml	=	0.137
M0D/Ml	=	0.290
N0DL/Ml	=	-0.074

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	52	51	52

Outplane Stress Summary	40	66	73	30
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	25	25	25	25

Inplane Stress Summary	20	29	36	18
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	40	66	73	30
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	14	41	48	2
-------------------------	----	----	----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	52	51	52

Inplane Stress Summary	45	56	61	45
------------------------	----	----	----	----



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		45	56	62	45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		1	1	1	1
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		0	0	0	0
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		8	9	8	9
Hoop Stress Summary		10	11	8	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		1	1	1	1
Axial Bending (P)		13	-13	13	-13
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-21	-21	21	21
Axial Bending (ML)		-61	61	61	-61
Axial Pressure Stress		4	4	4	4
Axial Stress Summary		-63	32	103	-48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		3	3	-3	-3
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		10	10	10	10
Shear Stress Summary		14	14	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		79	39	103	60

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9

Hoop Stress Summary	11	12	8	9
---------------------	----	----	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-62	62	62	-62
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-64	32	104	-49
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	14	14
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	76	34	106	65
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1655.5(18.0 - 3.0)}$$

$$= 634.325 \text{ mm}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 3311.000 \text{ mm}$$

$$\text{Nozzle Outside Diameter (d)} = 200.000 \text{ mm}$$

$$\text{Vessel Thickness used (T)} = 15.000 \text{ mm}$$

$$\text{Nozzle Thickness used (t)} = 49.585 \text{ mm}$$

$$T / t = 0.303$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.897$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$N_r/P = 0.111$$

$$M_r/P = 0.242$$

$$M_0/P = 0.070$$

$$N_0/P = 0.129$$

$$M_rD/Mc = 0.312$$

$$N_rDL/Mc = 0.073$$



$M\theta d/Mc = 0.096$
 $N\theta DL/Mc = 0.038$
 $MrD/Ml = 0.209$
 $NrDL/Ml = 0.091$
 $M\theta D/Ml = 0.066$
 $N\theta DL/Ml = 0.105$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	113	114	113	114
Outplane Stress Summary	108	117	135	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	44	67	118	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	117	136	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	48	66	82	37

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-51	51	51	-51
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	113	114	113	114
Inplane Stress Summary	84	142	192	43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	84	142	192	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	52	51	52	51	52	51	52
Circ. Pl (SUS)	1	1	1	1	-1	-1	1	1
Circ. Q (SUS)	-13	13	21	-21	-4	4	8	-8
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0
Long. Q (SUS)	-4	4	8	-8	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	52.0	51.0	52.0	51.0	52.0	51.0	52.0
Pm+Pl (SUS)	52.0	53.0	52.0	53.0	50.0	51.0	52.0	53.0
Pm+Pl+Q (Total)	39.2	66.1	73.1	32.3	46.1	55.3	60.3	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	73.10	413.99	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	113	114	113	114	113	114	113	114
Circ. Pl (SUS)	0	0	4	4	-1	-1	3	3
Circ. Q (SUS)	-4	4	16	-16	-28	28	74	-74
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	-1	-1	3	3	1	1	3	3
Long. Q (SUS)	-11	11	57	-57	-9	9	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	113.0	114.0	113.0	114.0	113.0	114.0	113.0	114.0
Pm+Pl (SUS)	113.0	114.0	117.0	118.0	112.0	113.0	116.0	117.0
Pm+Pl+Q (Total)	109.1	118.1	133.0	102.0	84.0	141.0	190.0	43.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	190.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-20	-20	22	22	-20	-20	22	22
Long. Q (SUS)	-48	48	74	-74	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	25.7	26.7	26.5	26.5	26.7	27.7	26.5	26.5
Pm+Pl+Q (Total)	77.5	38.0	100.5	59.7	76.3	35.0	102.8	63.6

Nozzle Stress Summation Comparison (MPa):

Expansion vessel (AMONIA PRODUCTION)

PV Elite 28 Licensee: SL User

FileName : A162-V-XX02-Expansion Vessel (Ammonia production) Page 135 of 181

Nozzle Calcs.: N5-4'' Nozl: 36 2:01pm Sep 17,2026



HEXAGON

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	27.66	177.00	Passed
Pm+Pl+Q (TOTAL)	102.78	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

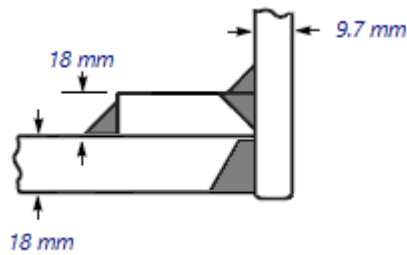
**Input, Nozzle Desc: N6-4''****From: 40**

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N6-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
 $= 12.4285 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.04 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
 $= 0.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618	mm
Parallel to Vessel Wall, opening length	d	100.8309	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.8364	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.855, 1.0)$$

$$= 0.855$$

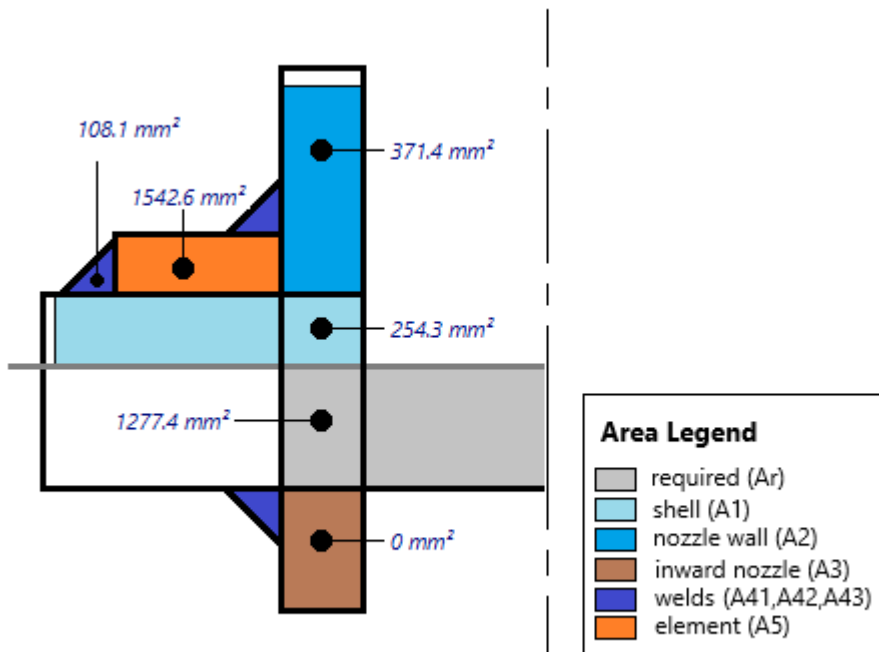


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1277.438	NA	NA
Area in Shell	A1	254.267	NA	NA
Area in Nozzle Wall	A2	371.423	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	108.082	NA	NA
Area in Element	A5	1542.600	NA	NA
TOTAL AREA AVAILABLE	Atot	2276.372	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	144.5036	18.0000 mm
Based on given Pad Diameter:	200.0000	6.3438 mm
Based on Shell or Nozzle Thickness:	170.1491	9.7345 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (100.8309 \cdot 12.4285 \cdot 1.0 + 2 \cdot 6.7345 \cdot 12.4285 \cdot 1.0 \cdot (1 - 0.86))$
 $= 1277.438 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 100.831(1.0 \cdot 15.0 - 1.0 \cdot 12.429) - 2 \cdot 6.735$
 $(1.0 \cdot 15.0 - 1.0 \cdot 12.4285) \cdot (1 - 0.855)$
 $= 254.267 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 34.84) \cdot (6.73 - 0.5) \cdot 0.8551$$

$$= 371.423 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (100.0) \cdot 0.86 + (0.0) \cdot 0.86 + 22.5746^2 \cdot 1.0$$

$$= 108.082 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot \text{fr4}$$

$$= (200.0 - 114.3) \cdot 18.0 \cdot 1.0$$

$$= 1542.600 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5000 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 15.4285 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 15.4285 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[8.258, \max(15.4285, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.4 MPa	Passed
Occasional	:	3.6,	Allowable	:	156.9 MPa	Passed
Shear	:	19.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.735, t_r = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, t_r = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, t_r = 0.5, c = 3.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.074, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 12.429, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.829$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.5, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.04/5.10 = 0.203$

Weld Size Calculations, Description: N6-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$7.5000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (1277.4377 - 254.267 + 2 \cdot 6.7345 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 12.4285)) \cdot 138)$
 $= 145269.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (371.4234 + 1542.6 + 108.0818 - 0.0 \cdot 0.86) \cdot 138$
 $= 279021.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (371.4234 + 0.0 + 85.5072 + (172.7558)) \cdot 138$
 $= 86887.78 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (371.4234 + 0.0 + 108.0818 + 1542.6 + (172.7558)) \cdot 138$
 $= 302859.59 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 103800. \text{ N}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.1416/2.0) \cdot 200.0 \cdot 14.0 \cdot 0.49 \cdot 138 \\
 &= 297378. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118 \\
 &= 93980. \text{ N}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138 \\
 &= 329993. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138 \\
 &= 274994. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (297378 + 93980) = 391358 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (103800 + 329993 + 274994 + 0) = 708788 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (297378 + 274994 + 0) = 572372 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 391357 N	, must exceed W = 145269 N	or W1 = 279021 N
Path 2-2 = 708787 N	, must exceed W = 145269 N	or W2 = 86887 N
Path 3-3 = 572371 N	, must exceed W = 145269 N	or W3 = 302859 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9930 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9930 mm

Input Echo, WRC297 Item 1, Description: N6-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3290.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.04	MPa
Radial Load	P	-3600.00	N
Circumferential Shear	Vc	4800.01	N
Longitudinal Shear	Vl	4800.01	N
Circumferential Moment	Mc	0.125E+07	N-mm
Longitudinal Moment	Ml	0.125E+07	N-mm
Torsional Moment	Mt	0.144E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1664.5(18.0 - 3.0))}$$

$$= 635.737 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3329.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	6.735	mm
T / t	=	4.900	
d / t	=	16.972	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.345$$

Nr/P	=	0.147
Mr/P	=	0.103
M0/P	=	0.246
N0/P	=	0.309
MrD/Mc	=	0.111
NrDL/Mc	=	0.135
M0d/Mc	=	0.300
N0DL/Mc	=	-0.136
MrD/Ml	=	0.111
NrDL/Ml	=	0.137
M0D/Ml	=	0.290
N0DL/Ml	=	-0.074

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	51	52	51	52

Outplane Stress Summary	40	66	73	30
-------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	25	25	25	25

Inplane Stress Summary	20	29	36	18
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	40	66	73	30
------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	14	41	48	2
-------------------------	----	----	----	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	52	51	52

Inplane Stress Summary	45	56	61	45
------------------------	----	----	----	----

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		45	56	62	45

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		1	1	1	1
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		0	0	0	0
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		8	9	8	9
Hoop Stress Summary		10	11	8	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		1	1	1	1
Axial Bending (P)		13	-13	13	-13
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-21	-21	21	21
Axial Bending (ML)		-61	61	61	-61
Axial Pressure Stress		4	4	4	4
Axial Stress Summary		-63	32	103	-48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		3	3	-3	-3
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		10	10	10	10
Shear Stress Summary		14	14	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		79	39	103	60

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9

Hoop Stress Summary	11	12	8	9
---------------------	----	----	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-62	62	62	-62
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-64	32	104	-49
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	14	14
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	76	34	106	65
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1655.5(18.0 - 3.0)}$$

$$= 634.325 \text{ mm}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 3311.000 \text{ mm}$$

$$\text{Nozzle Outside Diameter (d)} = 200.000 \text{ mm}$$

$$\text{Vessel Thickness used (T)} = 15.000 \text{ mm}$$

$$\text{Nozzle Thickness used (t)} = 49.585 \text{ mm}$$

$$T / t = 0.303$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.897$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$N_r/P = 0.111$$

$$M_r/P = 0.242$$

$$M_0/P = 0.070$$

$$N_0/P = 0.129$$

$$M_rD/Mc = 0.312$$

$$N_rDL/Mc = 0.073$$



$M\theta d/Mc = 0.096$
 $N\theta DL/Mc = 0.038$
 $MrD/Ml = 0.209$
 $NrDL/Ml = 0.091$
 $M\theta D/Ml = 0.066$
 $N\theta DL/Ml = 0.105$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	113	114	113	114
Outplane Stress Summary	108	117	135	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	44	67	118	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	117	136	101

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	48	66	82	37

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-51	51	51	-51
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	113	114	113	114
Inplane Stress Summary	84	142	192	43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	84	142	192	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	52	51	52	51	52	51	52
Circ. Pl (SUS)	1	1	1	1	-1	-1	1	1
Circ. Q (SUS)	-13	13	21	-21	-4	4	8	-8
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0
Long. Q (SUS)	-4	4	8	-8	-14	14	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	51.0	52.0	51.0	52.0	51.0	52.0	51.0	52.0
Pm+Pl (SUS)	52.0	53.0	52.0	53.0	50.0	51.0	52.0	53.0
Pm+Pl+Q (Total)	39.2	66.1	73.1	32.3	46.1	55.3	60.3	45.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	73.10	413.99	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	113	114	113	114	113	114	113	114
Circ. Pl (SUS)	0	0	4	4	-1	-1	3	3
Circ. Q (SUS)	-4	4	16	-16	-28	28	74	-74
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	-1	-1	3	3	1	1	3	3
Long. Q (SUS)	-11	11	57	-57	-9	9	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	113.0	114.0	113.0	114.0	113.0	114.0	113.0	114.0
Pm+Pl (SUS)	113.0	114.0	117.0	118.0	112.0	113.0	116.0	117.0
Pm+Pl+Q (Total)	109.1	118.1	133.0	102.0	84.0	141.0	190.0	43.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	138.00	Passed
Pm+Pl (SUS)	118.02	207.00	Passed
Pm+Pl+Q (TOTAL)	190.04	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-20	-20	22	22	-20	-20	22	22
Long. Q (SUS)	-48	48	74	-74	-49	49	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	25.7	26.7	26.5	26.5	26.7	27.7	26.5	26.5
Pm+Pl+Q (Total)	77.5	38.0	100.5	59.7	76.3	35.0	102.8	63.6

Nozzle Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	27.66	177.00	Passed
Pm+Pl+Q (TOTAL)	102.78	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K1A-2"

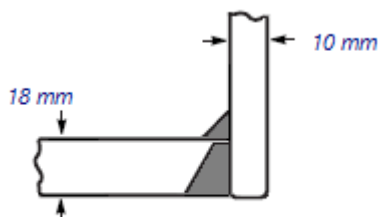
From: 40

Pressure for Reinforcement Calculations	P	1.0360	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----

Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.04 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.04)$
= 12.4285 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.04 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.04)$
= 0.2222 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+t	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2222	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.4285	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.4285	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(15.4285, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.2222, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, tg = 10.0, tr = 0.222, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.04 / 1.96 = 0.529$

Weld Size Calculations, Description: K1A-2''

Intermediate Calc. for nozzle/shell welds	Tmin	7.0000	mm
---	------	--------	----



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1962 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K1B-2"

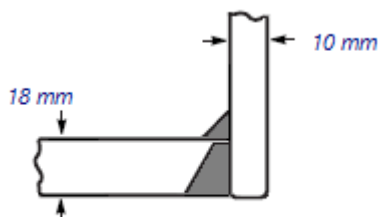
From: 40

Pressure for Reinforcement Calculations	P	1.0618	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000	in.
---	-------	-----



Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.06 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.06)$
= 12.7392 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $1.06 \cdot 25.4 / (118 \cdot 1.0 + 0.4 \cdot 1.06)$
= 0.2277 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+t	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-2''.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2277	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.7392	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.7392	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(15.7392, 4.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.2277, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, tg = 10.0, tr = 0.228, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.06 / 1.96 = 0.542$

Weld Size Calculations, Description: K1B-2''

Intermediate Calc. for nozzle/shell welds	Tmin	7.0000	mm
---	------	--------	----



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1961 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.1962 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

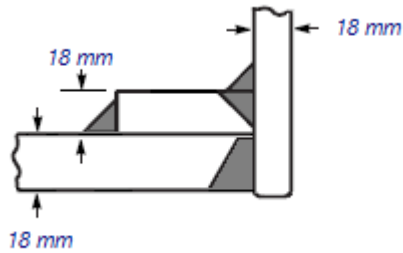
**Input, Nozzle Desc: N1-20''****From: 40**

Pressure for Reinforcement Calculations	P	1.0618	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3290.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6850.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		246.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.06 \cdot 1648.0 / (138 \cdot 1.0 - 0.6 \cdot 1.06)$
 $= 12.7392 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.06 \cdot 254.0 / (118 \cdot 1.0 + 0.4 \cdot 1.06)$
 $= 2.2774 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	956.0001 mm
Parallel to Vessel Wall, opening length	d	478.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

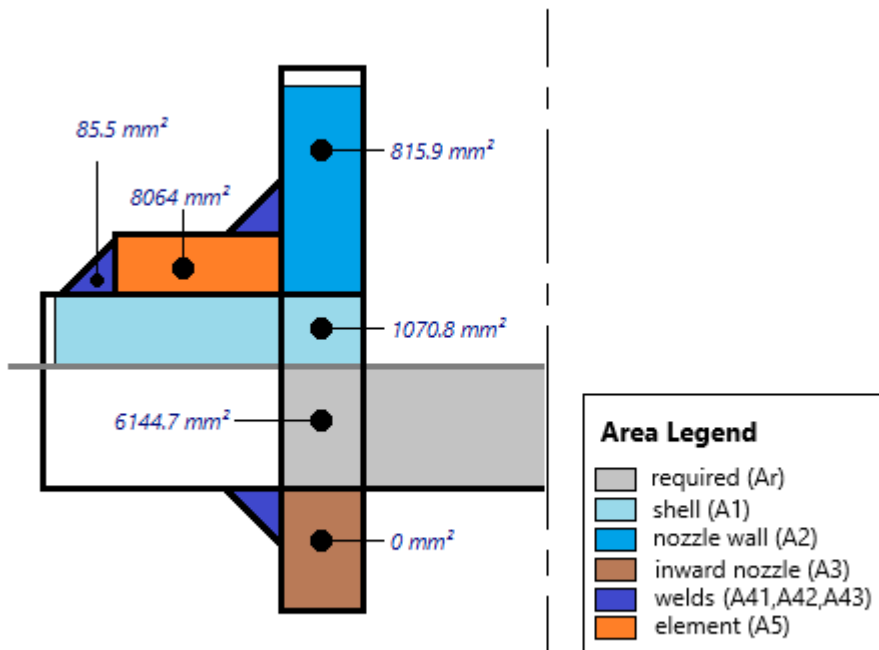


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6144.731	NA	NA
Area in Shell	A1	1070.828	NA	NA
Area in Nozzle Wall	A2	815.904	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	8064.001	NA	NA
TOTAL AREA AVAILABLE	Atot	10036.240	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	739.8051	18.0000 mm
Based on given Pad Diameter:	1000.0001	9.3136 mm
Based on Shell or Nozzle Thickness:	739.8051	18.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (478.0 \cdot 12.7392 \cdot 1.0 + 2 \cdot 15.0 \cdot 12.7392 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6144.731 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 478.0(1.0 \cdot 15.0 - 1.0 \cdot 12.739) - 2 \cdot 15.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 12.7392) \cdot (1 - 0.855)$
 $= 1070.828 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 37.5) \cdot (15.0 - 2.28) \cdot 0.8551$$
$$= 815.904 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$
$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$
$$= 85.507 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr4$$
$$= (956.0 - 508.0) \cdot 18.0 \cdot 1.0$$
$$= 8064.001 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.2774 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 15.7392 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr_{16b}) = 15.7392 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[11.331, \max(15.7392, 4.5)]$$
$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$
$$= \max(5.2774, 11.3312)$$
$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	25.7,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	269.3 MPa	Passed
Occasional	:	8.2,	Allowable	:	156.9 MPa	Passed
Shear	:	12.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 18.0$, $t_r = 2.277$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.152$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 2.277$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.152$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 2.277$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.152$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 12.739, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.849$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -19 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 12.739, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.849$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -19 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -29 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.06/1.96 = 0.542$

Weld Size Calculations, Description: N1-20"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (6144.731 - 1070.8284 + 2 \cdot 15.0 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 12.7392)) \cdot 138)$
 $= 708128.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (815.9036 + 8064.0015 + 85.5072 - 0.0 \cdot 0.86) \cdot 138$
 $= 1237099.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (815.9036 + 0.0 + 85.5072 + (384.7826)) \cdot 138$
 $= 177476.39 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (815.9036 + 0.0 + 85.5072 + 8064.0015 + (384.7826)) \cdot 138$
 $= 1290194.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 461336. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1000.0 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 1486888. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 246.5) \cdot (18.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 959386. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 1466635. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 1222196. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1486888 + 959386) = 2446274 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (461336 + 1466635 + 1222196 + 0) = 3150167 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (1486888 + 1222196 + 0) = 2709085 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2446274 N	, must exceed W = 708128 N	or W1 = 1237099 N
Path 2-2 = 3150167 N	, must exceed W = 708128 N	or W2 = 177476 N
Path 3-3 = 2709084 N	, must exceed W = 708128 N	or W3 = 1290194 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 19.7280 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 437.7280 mm

Input Echo, WRC297 Item 1, Description: N1-20" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3290.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	18.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1000.000	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.06	MPa
Radial Load	P	-31998.60	N
Circumferential Shear	Vc	23998.92	N
Longitudinal Shear	Vl	31998.60	N
Circumferential Moment	Mc	0.320E+08	N-mm
Longitudinal Moment	Ml	0.416E+08	N-mm
Torsional Moment	Mt	0.480E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(1664.5(18.0 - 3.0))}$$

$$= 1029.437 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3329.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	15.000	mm
T / t	=	2.200	
d / t	=	33.867	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.533$$

Nr/P	=	0.082
Mr/P	=	0.097
M0/P	=	0.062
N0/P	=	0.171
MrD/Mc	=	0.190
NrDL/Mc	=	0.112
M0d/Mc	=	0.141
N0DL/Mc	=	0.170
MrD/Ml	=	0.123
NrDL/Ml	=	0.080
M0D/Ml	=	0.082
N0DL/Ml	=	0.298

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	5	5	5	5
------------------------	---	---	---	---



Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	52	53	52	53

Outplane Stress Summary	9	62	127	33
-------------------------	---	----	-----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	-15	60	107	-38
------------------------	-----	----	-----	-----

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	26	66	128	71

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	-16	59	100	-18
-------------------------	-----	----	-----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-66	66	66	-66
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53



Inplane Stress Summary	0	98	144	-20
------------------------	---	----	-----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	16	98	145	24
------------------------	----	----	-----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20

Hoop Stress Summary	2	3	47	48
---------------------	---	---	----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-181	181	181	-181
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-136	129	255	-203
----------------------	------	-----	-----	------

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
----------------------	---	---	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	140	130	256	252
------------------------	-----	-----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20
Hoop Stress Summary	14	16	34	35

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	-11	-11	11	11
Axial Bending (Mc)	-225	225	225	-225
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-177	177	296	-251

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7
Shear Stress Summary	5	5	10	10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	192	177	297	287

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(1655.5(18.0 - 3.0))}$$

$$= 1028.025 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3311.000 mm
Nozzle Outside Diameter	(d) =	1000.000 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	261.000 mm
T / t	=	0.057
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.487$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.076
M0/P	=	0.021
N0/P	=	0.025



MrD/Mc = 0.245
 NrDL/Mc = 0.106
 M0d/Mc = 0.073
 N0DL/Mc = 0.049
 MrD/Ml = 0.053
 NrDL/Ml = 0.057
 M0D/Ml = 0.016
 N0DL/Ml = 0.027

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	116	117	116	117
Outplane Stress Summary	114	115	160	90

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	60	48	198	-47

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	114	115	198	137

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary 10 99 148 -11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-209	209	209	-209
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117

Inplane Stress Summary -36 253 411 -134

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 0 0 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 46 253 412 134

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	-17	-17	27	27	-4	-4	8	8
Circ. Q (SUS)	-25	25	47	-47	-49	49	83	-83
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-4	-4	8	8	-4	-4	14	14
Long. Q (SUS)	-38	38	72	-72	-37	37	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	35.0	36.0	79.0	80.0	48.0	49.0	60.0	61.0
Pm+Pl+Q (Total)	26.7	63.5	126.4	71.3	15.3	98.1	143.4	24.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed



Pm+Pl (SUS)	80.00	207.00	Passed
Pm+Pl+Q (TOTAL)	143.36	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	-1	-1	7	7	-8	-8	22	22
Circ. Q (SUS)	0	0	34	-34	-145	145	273	-273
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-3	-3	17	17	-3	-3	9	9
Long. Q (SUS)	6	-6	122	-122	-45	45	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+Pl (SUS)	115.0	116.0	123.0	124.0	108.0	109.0	138.0	139.0
Pm+Pl+Q (Total)	115.2	116.1	197.0	137.0	47.0	254.0	411.0	134.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+Pl (SUS)	139.01	207.00	Passed
Pm+Pl+Q (TOTAL)	411.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-17	-17	27	27	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-13	-13	15	15	-10	-10	12	12
Long. Q (SUS)	-133	133	229	-229	-177	177	273	-273
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	6.3	7.3	46.0	47.0	16.5	17.5	33.3	34.3
Pm+Pl+Q (Total)	139.9	129.5	253.2	252.3	193.3	176.2	294.3	286.6

**Nozzle Stress Summation Comparison (MPa):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	118.00	Passed
Pm+Pl (SUS)	47.04	177.00	Passed
Pm+Pl+Q (TOTAL)	294.31	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N2-4''

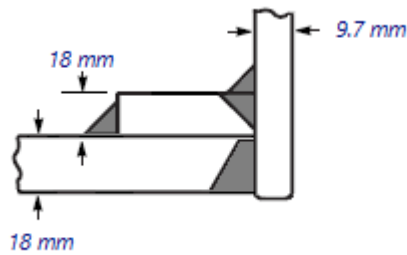
Parent: N1-20''

Pressure for Reinforcement Calculations	P	1.0618	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	472.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.06 \cdot 239.0 / (118 \cdot 1.0 - 0.6 \cdot 1.06)$
 $= 2.1623 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.06 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 1.06)$
 $= 0.5124 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	201.6618 mm
Parallel to Vessel Wall, opening length	d	100.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.8364 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

$$\text{Area Available in Welds [A41 + A42 + A43]:}$$

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$



$$= (100.0) \cdot 1.0 + (0.0) \cdot 1.0 + 22.5746^2 \cdot 1.0$$

$$= 122.575 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (200.0 - 114.3) \cdot 18.0 \cdot 1.0$$

$$= 1542.600 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.5124 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 5.1623 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 5.1623 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.258, \max(5.1623, 4.5)]$$

$$= 5.1623 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5124, 5.1623)$$

$$= 5.1623 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.735$, $tr = 0.512$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.735$, $tr = 0.512$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.735$, $tr = 0.512$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 2.162$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.144$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.735$, $tr = 0.512$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -45 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1.06/1.96 = 0.542

Weld Size Calculations, Description: N2-4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.7142 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (218.0296 - 1294.434 + 2 • 6.7345 • 1.0 •
(1.0 • 15.0 - 2.1623))118)
= max(0, -106601.12) N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (433.5127 + 1542.6 + 122.5746 - 0.0 • 1.0) • 118
= 247619.56 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (433.5127 + 0.0 + 100.0 + (202.0365)) • 118
= 86785.86 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (433.5127 + 0.0 + 122.5746 + 1542.6 + (202.0365)) • 118
= 271457.44 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 114.3 • 10.0 • 0.49 • 118
= 103800. N

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.1416/2.0) • 200.0 • 14.0 • 0.49 • 118
= 254279. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 53.7827) • (9.7345 - 3.0) • 0.7 • 118
= 93980. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg



$$= (3.1416/2) \cdot 114.3 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 329993. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 118$$

$$= 235140. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (254279 + 93980) = 348260 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103800 + 329993 + 235140 + 0) = 668933 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (254279 + 235140 + 0) = 489419 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 348259 N	, must exceed W = 0 N	or W1 = 247619 N
Path 2-2 = 668933 N	, must exceed W = 0 N	or W2 = 86785 N
Path 3-3 = 489419 N	, must exceed W = 0 N	or W3 = 271457 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.0243 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.0242 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M2-24''

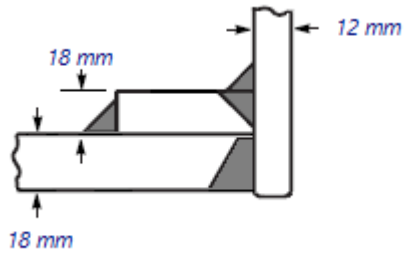
From: 50

Pressure for Reinforcement Calculations	P	1.0457	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3290.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.05 \cdot 3296.0 \cdot 0.8984 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.05)$
 $= 11.2280 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.05 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 1.05)$
 $= 2.3027 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001	mm
Parallel to Vessel Wall, opening length	d	591.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

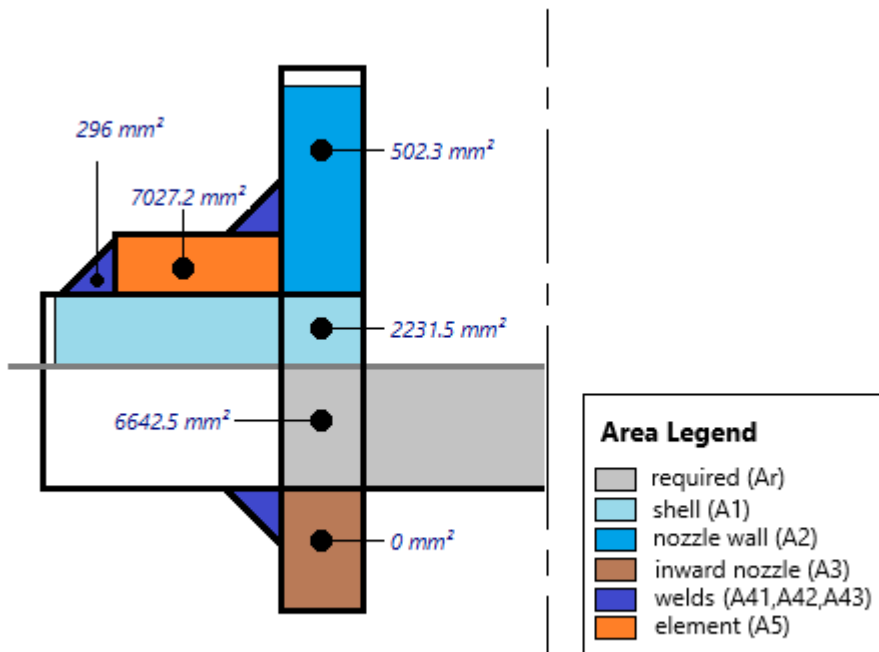


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6642.510	NA	NA
Area in Shell	A1	2231.491	NA	NA
Area in Nozzle Wall	A2	502.300	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	296.000	NA	NA
Area in Element	A5	7027.200	NA	NA
TOTAL AREA AVAILABLE	Atot	10056.991	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	810.3065	18.0000 mm
Based on given Pad Diameter:	1000.0000	9.2539 mm
Based on Shell or Nozzle Thickness:	910.6599	12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (591.6 \cdot 11.228 \cdot 1.0 + 2 \cdot 9.0 \cdot 11.228 \cdot 1.0 \cdot (1 - 1.0))$
 $= 6642.510 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 591.6(1.0 \cdot 15.0 - 1.0 \cdot 11.228) - 2 \cdot 9.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 11.228) \cdot (1 - 1.0)$
 $= 2231.491 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 37.5) \cdot (9.0 - 2.3) \cdot 1.0$$

$$= 502.300 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$$

$$= 296.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (1000.0 - 609.6) \cdot 18.0 \cdot 1.0$$

$$= 7027.200 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.3027 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.3027 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 11.228$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.749$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.303$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.256$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1.05/1.96 = 0.533

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	8.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (6642.5098 - 2231.491 + 2 • 9.0 • 1.0 •
(1.0 • 15.0 - 11.228))138)
= 618026.38 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (502.3004 + 7027.2002 + 296.0 - 0.0 • 1.0) • 138
= 1079807.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (502.3004 + 0.0 + 100.0 + (270.0)) • 138
= 120365.03 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (502.3004 + 0.0 + 296.0 + 7027.2002 + (270.0)) • 138
= 1117063.88 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 609.6 • 10.0 • 0.49 • 138
= 647434. N

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.1416/2.0) • 1000.0 • 14.0 • 0.49 • 138
= 1486888. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 300.3) • (12.0 - 3.0) • 0.7 • 138
= 820125. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg
= (3.1416/2) • 609.6 • 18.0 • 0.74 • 138
= 1759962. N



Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$
= $(3.1416/2.0) \cdot 609.6 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$
= 1466635. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SPEW + SNW) = (1486888 + 820125) = 2307014 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (647434 + 1759962 + 1466635 + 0) = 3874031 N

Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (1486888 + 1466635 + 0) = 2953524 N

Summary of Failure Path Calculations:

Path 1-1 = 2307013 N	, must exceed W = 618026 N	or W1 = 1079807 N
Path 2-2 = 3874031 N	, must exceed W = 618026 N	or W2 = 120365 N
Path 3-3 = 2953523 N	, must exceed W = 618026 N	or W3 = 1117063 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 15.6529 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 333.6530 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.008 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.